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Community Twin Ecosystem for Disaster Resilient Communities

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Highlights:

What are the main findings?

- COWINE captures the intricate, multidimensional, and interconnected dynamics of communities leveraging Digital Twin technology, allowing for detailed disaster resilience planning and collaboration among stakeholders.
- The case study in Brevard County, Florida, demonstrates COWINE's capability as a collaborative Digital Twin-based ecosystem to identify vulnerable areas and aid in the execution of preventive and adaptive strategies in order to enhance resource allocation efficiency before, during, and after a disaster.

What is the implication of the main finding?

- COWINE's ability to model complex urban dynamics and support decision-making through stakeholder collaboration highlights its potential to transform community disaster resilience management, offering a scalable and transferable approach to proactive disaster planning and response.
- The use of such Digital Twin-based approaches or ecosystems like COWIN^{E} in community resilience management is pivotal in enabling informed decision-making in the event of disasters, particularly in today's world of frequent and increasingly severe natural hazards.

Abstract: This paper presents COWIN^{E} (Community Twin Ecosystem), an ecosystem that harnesses Digital Twin (DT) to elevate and transform community resilience strategies. COWIN^{E} aims to enhance the disaster resilience of communities by fostering collaborative participation in the use of its DT among the decision-makers, the general public, and other involved stakeholders. COWIN^{E} leverages Cities:Skylines as its base simulation engine integrated with real-world data for community DT development. It is capable of capturing the dynamic, intricate, and interconnected structures of communities to provide actionable insights into disaster resilience planning. Through demonstrative, simulation-based case studies on Brevard County, Florida, the paper illustrates COWIN^{E}'s collaborative use with the involved parties in managing tornado scenarios. This study demonstrates how COWIN^{E} supports the identification of vulnerable areas, the execution of adaptive strategies, and the efficient allocation of resources before, during, and after a disaster. This paper further explores potential research directions using COWIN^{E}. The findings show COWIN^{E}'s potential to be utilized as a collaborative tool for community disaster resilience management.

Keywords: community resilience; digital twin; disaster management; infrastructure resilience; urban dynamics; social vulnerability

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1. Introduction

Prioritizing disaster preparedness has become crucial for supporting community resilience [1,2]. Community resilience management involves systematic efforts to strengthen a community's ability to withstand, adapt to, and recover from disasters, which is a core objective of resilience initiatives [3]. As societies confront adversities like earthquakes, hurricanes, economic downturns, and public health crises, enhancing community resilience and effective emergency management is increasingly emphasized in policy, research, and practice [4,5]. This holistic view recognizes resilience as a multifaceted concept requiring an understanding of the intricate interplay among individual actions, community institutions, global influences, and technological advancements [6,7].

1.1. Current Challenges in Community Resilience

Achieving disaster resilience in communities (community/city/urban used interchangeably in this paper) is a crucial goal, but significant challenges exist, hindering its effective implementation [8]. Insights gained from our extensive interviews and discussions with leaders from non-governmental organizations, as well as county climate and sustainability officers/engineers, reveal critical barriers to this goal. Our findings highlight significant challenges in realizing community resilience [9,10], presented in the following.

(1) There is a critical need for a theoretical understanding of disaster learning mechanisms that characterize a relationship between communities' past disaster experiences, learning from them, and using them to face future disasters with greater built-up resilience capacity;
(2) Advancing community resilience requires gaining further insight into the ever-changing interconnections among different facets of communities. Communities are dynamic, intricate systems where social, economic, environmental, and infrastructural elements are intertwined;
(3) During disasters, socio-economically disadvantaged communities experience disproportionate levels of distress. These communities often lack the resources and infrastructure needed to respond to and recover from disasters effectively. Residents of these communities tend to rely on local organizations and leaders for support;
(4) Households, emergency managers, and decision-makers typically may not know the most critical and vulnerable parts of the communities when faced with natural disasters. This lack of accurate vulnerability assessment can lead to the ineffective allocation of resources and efforts in disaster preparedness and response;
(5) Evacuation models are not robust enough to handle unpredictable, changing environments, particularly when reaching out to the public. Existing models may fail to account for the complex and dynamic nature of real-world scenarios, leading to suboptimal evacuation strategies that can endanger lives and hinder efficient disaster management;
(6) It is also important to recognize the growing need for transferable, scalable, and sustainable [11], as well as intelligent, community resiliency frameworks. These must be capable of encompassing the intricate dynamics of different communities to estimate their resilience performance against unpredictable, varying disturbances [12,13].

Addressing these challenges and other similar difficulties faced in community resilience practices [9,10] requires systematic approaches that encompass communities' multifaceted, dynamic dimensions. Digital Twin (DT) stands out as one of the most promising tools among the developments in today's world [14], possessing great potential to address the pressing challenges in achieving disaster resilience in communities. DT's capability of replication, thus enabling the management of complex systems in a controlled digital environment, provides high-level control over the constructed/designed system and facilitates proactive decision-making through what-if scenario planning and predictive analytics [15]. This paper introduces COWIN^{E} (Community Twin Ecosystem). COWIN^{E}, as illustrated in Figure 1, offers sophisticated simulation capabilities and actionable insights to provide a

transformative approach to community resilience practices. While COWIN^{E} is designed to address some of these resilience challenges, as mentioned earlier, this paper's scope is limited to presenting COWIN^{E}, and subsequent studies will investigate experimental works to address them.

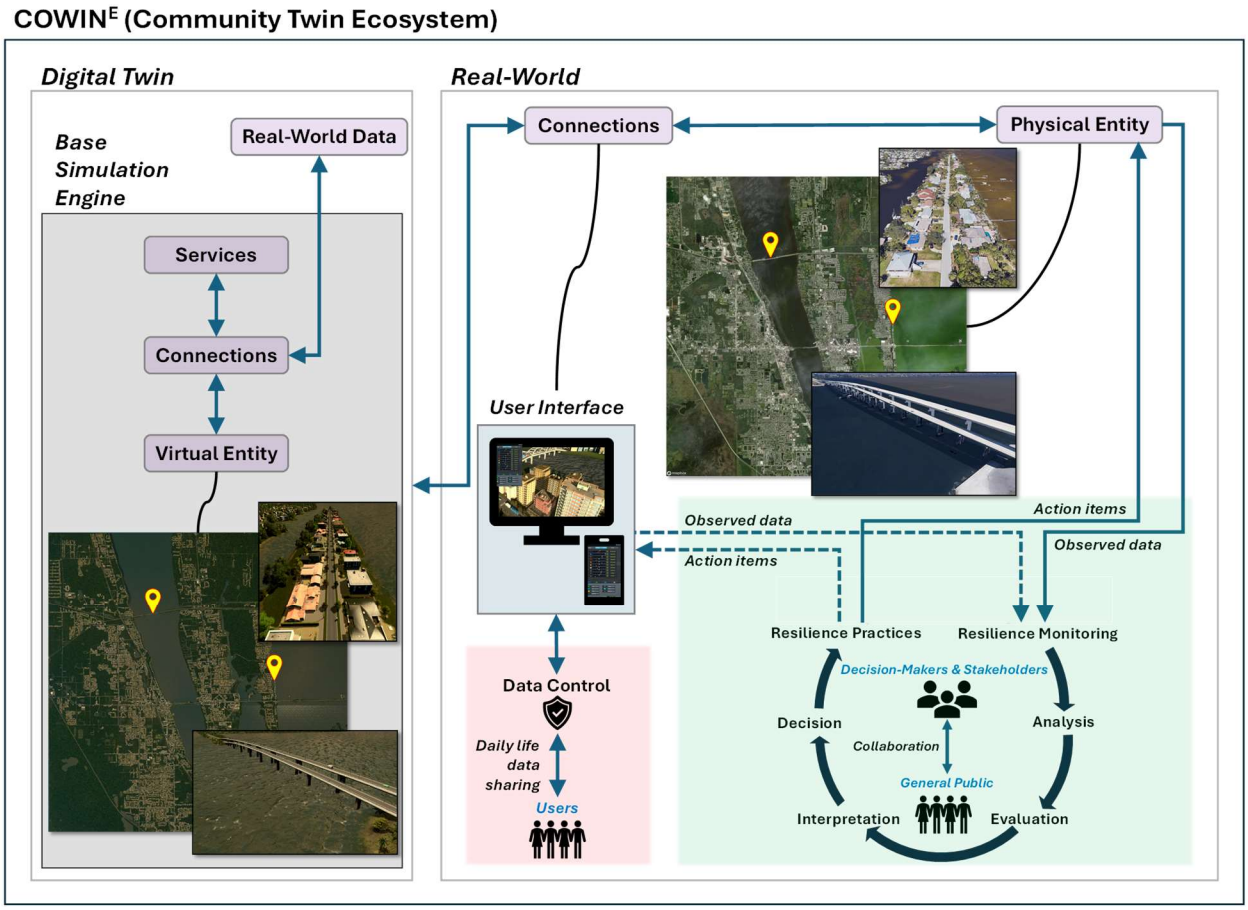


Figure 1. Graphical abstract: Community Twin Ecosystem (COWIN^{E}) showcasing its components with interactions. For the observed data and action items lines, the dashed line represents the interaction of decision-makers & stakeholders, and the public with the DT's user interface; the solid line represents the interaction with the physical entity.

1.2. Knowledge Gaps and Study Contribution

The current landscape of community resilience research reveals several limitations that this study seeks to address. The literature background regarding these limitations is provided in the next section, and a brief overview of these limitations is outlined below:

- While current resilience frameworks provide valuable tools for assessing community resilience, they predominantly rely on static models that focus on input–output relationships. Such models fail to capture the intricate, dynamic responses of communities to disturbances, as real-world communities often react in complex, nonlinear ways. This limitation restricts the applicability of these frameworks in practical scenarios where adaptive capacities and community interactions play critical roles in resilience-building.
- DT technology has emerged as a powerful tool with the potential to address these challenges through dynamic, virtual representations of physical systems. However, most DT studies remain conceptual, focusing on high-level frameworks without demonstrating the functional capabilities necessary for practical application in community resilience.

- Existing DT studies tend to concentrate on isolated dimensions or components, overlooking the interconnected and multifaceted characteristics of communities. Without accounting for the interwoven social, economic, environmental, and infrastructural aspects, these fragmented approaches provide an incomplete understanding of community resilience, which limits their effectiveness in real-world resilience planning and response.
- The visualization components in current DT applications are often static and lack interactivity, which reduces their effectiveness for stakeholders. This limitation impacts engagement from decision-makers, the public, and other stakeholders, as static and unintuitive visualizations hinder user experience, understanding, and practical usability.

This paper addresses the limitations of current community resilience research by introducing COWIN^{E} (Community Twin Ecosystem), an ecosystem that leverages DT technology to enhance community resilience strategies. Built on the base simulation engine Cities:Skylines (C:S) and integrated with real-world data, COWIN^{E} aims to foster disaster resilience in communities. It achieves this through active, collaborative engagement among the general public, decision-makers, and other stakeholders during disaster events. The primary contributions of COWIN^{E} are that it provides a practical and comprehensive digital representation of communities, capable of capturing the dynamic, intricate, multidimensional, and interconnected nature of community systems. Moving beyond traditional static models, COWIN^{E} features interactive DT visualizations that enable stakeholders to simulate various scenarios and collaboratively enhance decision-making capabilities. Additionally, the platform features a user-friendly and intuitive interface, promoting broad accessibility and usability, making it a valuable tool for real-time community resilience planning and response. Overall, considering the existing literature, as given in the next section, COWIN^{E} represents a considerable advancement in the application of DT technology for enhancing community resilience and addressing the complexities of disaster management.

2. Background and Review

2.1. Community Resilience

Numerous studies have explored resilience at various scales, contexts, and in response to different disturbances [16–24]. Bruneau et al. [25] developed a framework to measure the seismic resilience of communities, categorizing resilience into dimensions, properties, and consequences [26–28]. Building on this, the Hyogo Framework for Action [29] was established to enhance national resilience through government and policy-level measures [30], and this was further expanded by Kammouh et al. [31]. Renschler et al. [32] introduced the PEOPLES resilience structure, integrating seven key dimensions for assessing community resilience: population and demographics, environment/ecosystem, organized governmental services, physical infrastructure, lifestyle and community competence, economic development, and social–cultural capital [33]. Complementing these frameworks, FEMA's Hazus-MH [34] offers a risk modeling tool to estimate damage from natural disasters, available as downloadable software using ArcGIS desktop [35–38].

The BRIC (Baseline Resilience Indicators for Communities) index [39] evaluates disaster resilience at the county level in the United States, measuring various aspects across six categories to facilitate comparisons and track resilience improvements over time. In parallel, FEMA's National Risk Index (NRI) [40] utilizes this BRIC dataset to identify community-level risks and resilience. NIST (National Institute of Standards and Technology) developed the Economic Decision Guide Software (EDGe$) [41] to assist communities in making informed decisions about resilience projects through economic analysis, targeting professionals involved in cost-efficient resilience planning. NIST also supports the Center of Excellence for Risk-Based Community Resilience Planning, which created IN-CORE (Interdependent Networked Community Resilience Modeling Environment) to enable quantitative comparisons of resilience strategies by integrating physical and socioeconomic models [42]. Moreover, Ellingwood et al. [43] developed the Centerville Virtual

Community, serving as a testbed for community resilience and forming the foundation for a comprehensive decision framework [44–47].

In the United States, the Department of Homeland Security addresses critical infrastructure and community resilience, providing strategic guidance through programs within the Office of Infrastructure Protection and FEMA's National Preparedness Goal and Framework [48–50]. In Europe, the Joint Research Center's Geospatial Risk and Resilience Assessment Platform [51] uses geospatial technologies and computational tools to analyze and simulate critical infrastructure resilience [52–54]. Globally, initiatives such as the United Nations' International Strategy for Disaster Reduction Resilience Scorecard [55] and the Rockefeller Foundation's 100 Resilient Cities program [56], launched in 2013, aim to help communities worldwide address various challenges, including disasters, unemployment, food insecurity, housing affordability, water shortages, and transportation issues [57]. Several other efforts by communities, states, non-profit organizations, and researchers also contribute to enhancing community resilience through the development of guidance and assessment methodologies. Examples include SPUR (San Francisco Bay Area Planning and Urban Research Association) [58], the Coastal Resilience Index by the National Oceanic and Atmospheric Administration [59], the Community and Regional Resilience Institute (CARRI) [60], the Oregon Resilience Plan [61], and ResilUS [62].

While these endeavors provide valuable frameworks and tools for assessing community resilience, they predominantly rely on static models that focus on input–output relationships. Such models fail to capture the intricate and dynamic nature of communities as the communities respond to disturbances in complex, nonlinear ways. These works fall short of offering a comprehensive understanding of community disaster resilience, limiting their applicability in real-world scenarios where adaptive capacities and interactions play crucial roles in resilience-building efforts.

In contrast, advancements in technological tools have accompanied the evolving discourse on community resilience practices. DT has been one pivotal advancement with the potential to address these challenges by offering a dynamic virtual representation of physical systems. Integrating DTs with established community resilience frameworks can potentially create a synergistic approach that enhances decision-making capabilities through improved monitoring and what-if-like simulations, hence providing a deeper understanding of resilience. This integration ultimately fosters more resilient responses to a variety of disturbances in communities. A few studies have explored the DT in community resilience contexts, which are presented in the following sections. However, it is pertinent to introduce DT first.

2.2. Digital Twin (DT)

The term Digital Twin (DT) was introduced to the public by Michael Grieves in a product life-cycle management course in 2003 [63]. Since then, it has also been referenced under different terms, e.g., virtual twins, and later appeared in a paper by NASA in 2012 as "Digital Twin" [64]. NASA was one of the early adopters of DT technology for spacecraft and systems management, as the Apollo program's virtual simulation models are often considered the precursors to modern DTs. The earlier applications of DT also emerged in manufacturing sectors, where companies began using DTs to enable the real-time monitoring, diagnostics, and prognostics of equipment and processes [65]. In the structural engineering field, the subject of model updating has played a significant role in the collective development of DT. Model updating is a fundamental process for calibrating the virtual system based on data from the physical system to ensure higher accuracy in simulations and has been the focus of extensive research [66–68].

DT was first presented conceptually based on three main components: the Physical Entity, the Virtual Entity, and the data connections that link them [63]. Over time, the three dimensions of DT were extended to five dimensions to meet the new requirements of DT modeling, adding data and services [69,70]. These five dimensions of DT could also be extended to further dimensions based on future needs [71]. COWIN^{E}'s DT employs the

five-dimensional DT concept, which is provided in Figure 2, considering the DT system of a city environment.

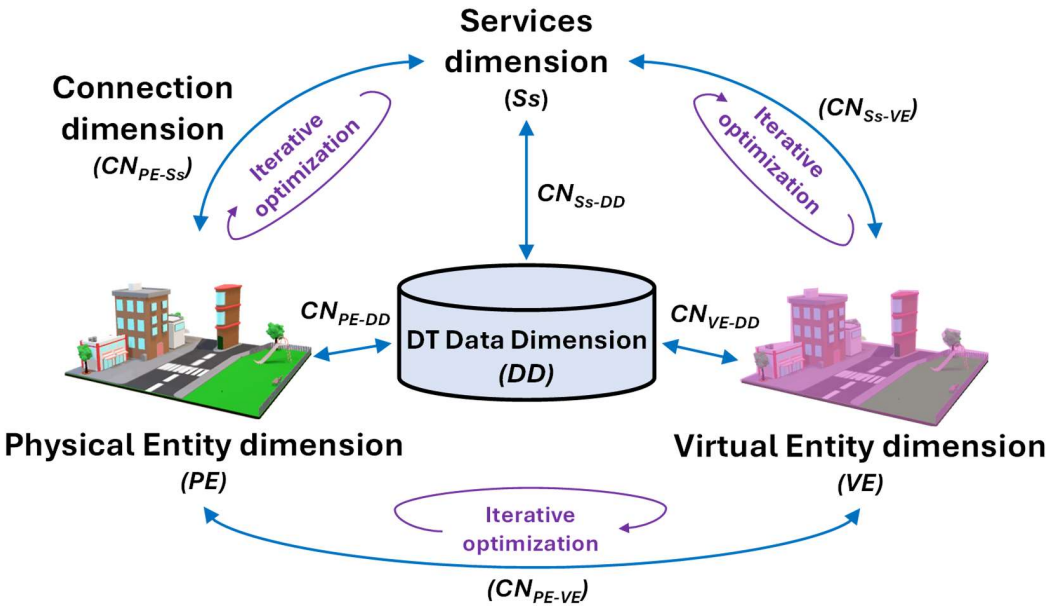


Figure 2. Five-dimensional DT structure: *DD* is the DT Data, *PE* is the Physical Entity, *VE* is the Virtual Entity, *Ss* is the Services, and $CN_{PE\text{-}Ss/PE\text{-}DD/PE\text{-}VE/Ss\text{-}DD/VE\text{-}DD/Ss\text{-}VE}$ is the Connection dimensions.

The Physical Entity (*PE*) dimension represents the actual physical components of the city, including infrastructure such as roads, buildings, utilities, and public services, as well as sensory devices that monitor environmental and operational conditions across the urban environment. The Virtual Entity (*VE*) dimension is a high-fidelity digital representation of the city, integrating parameters such as the city's geometry *Gv* (e.g., the 3D spatial layout of the city, encompassing models of buildings, roads, bridges, utilities, and the natural topography, including elevation, water bodies, and green spaces), physical properties *Pv* (e.g., the characteristics and attributes of the city's components, such as the material properties of buildings and infrastructure, environmental factors like climate and weather conditions, and the capacities of utilities and services), behaviors *Bv* (e.g., the dynamic activities and interactions within the city, including traffic flow, pedestrian movement, energy consumption, waste management, and the response of the city's systems to various events, such as emergencies or peak usage times), and operational rules *Rv* (e.g., the policies, regulations, and control mechanisms that govern the city's operations, such as zoning laws, traffic regulations, building codes, and emergency response protocols, which ensure the city functions efficiently and safely), enabling the simulation of urban dynamics, infrastructure performance, and citizen behavior in a virtual environment. The *VE* dimension is characterized by Equation (1).

$$VE \cong (Gv, Pv, Bv, Rv) \tag{1}$$

The Services (*Ss*) dimension includes a range of functions that optimize operations in the *PE* dimension, ensure the high fidelity of the *VE* dimension by calibrating its parameters during operation, and provide ongoing support for monitoring, reporting, predictive analytics, scenario simulation, and decision support in the DT [72,73]. These services maintain alignment between the *PE* and *VE*, enhance the overall performance and management of the city's systems, and provide decision-making services. They are described by their *Function*, *Input*, *Output*, *Quality*, and *State*, as given in Equation (2).

$$Ss \cong (Function, Input, Output, Quality, State) \tag{2}$$

Services can be scheduled and adjusted to meet the evolving demands of both the physical city and its twin. For example, consider an infrastructure health monitoring service for the city via Equation (2). This service could be represented as *Ss_infrastructure_health = (Infrastructure health monitoring, {real-time data from structural sensors, simulated stress and wear data from VE}, infrastructure condition, {response time, accuracy, reliability}, {healthy, at risk, critical})*. Here, the function is infrastructure health monitoring; the inputs are real-time data from sensors monitoring the physical infrastructure and simulated stress and wear data from the virtual city; the output is the current infrastructure condition; the quality is assessed by response time, accuracy, and reliability; and the state reflects whether the infrastructure is healthy, at risk, or in a critical condition [74]. This service ensures that the virtual model accurately reflects the real-world condition of the city's infrastructure and provides timely insights for maintenance and risk management.

The DT Data (*DD*) dimension stores the real-time data flow between the other DT dimensions, capturing the current state and performance of the city. In this structure, *Dp* represents the data collected from the *PE* dimension of the city, such as sensor readings from traffic, environmental monitoring, and infrastructure health [75]. *Dv* refers to data in the *VE* dimension, including outputs from simulations and models that predict urban dynamics like traffic patterns and energy consumption. *Ds* is the data associated with the *Ss* dimension, which, for example, in the case of *Ss_infrastructure_health*, would include inputs like real-time data from structural sensors, outputs like infrastructure condition assessments, and quality metrics such as the response time, accuracy, and reliability of the service. These data are critical for monitoring the efficiency and effectiveness of city services, including emergency response times, public transportation schedules, and utility management. *Dk* represents domain-specific knowledge, incorporating specialized information such as urban planning guidelines, historical data, and regulatory frameworks. Lastly, *Df* is the fusion of these data sources [76], integrating *Dp*, *Dv*, *Ds*, and *Dk* to create a comprehensive dataset that provides a holistic view of the city's operations and supports accurate simulation and decision-making within the DT. The *DD* dimension is characterized by Equation (3).

$$DD \cong (Dp, Dv, Ds, Dk, Df) \tag{3}$$

The Connection (*CN*) dimension includes the bidirectional communication infrastructure that facilitates interaction and data exchange among the *PE*, *VE*, *DD*, and *Ss*, ensuring synchronization and coherence across the DT, with specific connections such as $CN_{Ss\text{-}DD}$, $CN_{PE\text{-}DD}$, $CN_{VE\text{-}DD}$, $CN_{PE\text{-}Ss}$, $CN_{VE\text{-}Ss}$, and $CN_{PE\text{-}VE}$. These connections enable the continuous (ideally real-time) flow of data, allowing the *PE* dimension to inform the *VE* dimension, the *SS* dimension to be calibrated and optimized in real time, and all components to be accurately reflected in the *DD* dimension. Each connection is characterized in Equation (4):

$$CN_XX \cong (Datasource, Unit, Value, Scope, Samplinginterval) \tag{4}$$

where Data source would be the sensors installed on critical infrastructure, such as a bridge, which monitor structural conditions [77]; the Unit might be millimeters (mm) for displacement; the Value represents the actual reading from the sensor, a displacement of 2 mm; the Scope outlines the acceptable range of these values, for example, a displacement range of 0–5 mm; and the Sampling interval specifies how often these data are collected and transmitted, e.g., every tenth of a second for the continuous monitoring of structural health.

2.3. Digital Twin in Community Resilience

Developing a DT of designed/constructed systems has been the ideal approach for an optimized system life-cycle [78]. This approach has influenced major fields such as civil engineering and has led to many works on various systems and aspects of the built environment [71,79–81]. As such, some works have explored DTs for critical infrastructure networks [82–85], including an investigation of the requirements and challenges associated with integrating monitoring systems into digital twin platforms [85]. Some other studies

explored the flood risks and water infrastructure impacts during epidemics by developing DTs [86–88]. Agapaki studied the DT implementation of airports in the context of resilient disaster response [89]. Braik and Koliou developed a DT of the Galveston testbed's electric power infrastructure systems to study the infrastructure system subjected to hurricanes [90].

In addition to investigating the applications of DTs at the component level of the built environment, such as bridges, airports, and transmission towers, researchers have also explored their broader-level applications and encompassed entire communities and cities while considering the interconnected dynamics of various regions to enhance resilience practices. Ye et al. [91] reviewed urban DT studies, identifying challenges and opportunities for community adaptation and proposing a human-centered framework to enhance resilience through multi-agent interactions and artificial intelligence integration. Lagap and Ghaffarian [92] analyzed the potential of DTs in disaster risk management, proposing an improved framework for post-disaster response and recovery. Ariyachandra and Wedawatta [93] explored the evolving concepts of DT smart cities for disaster risk management, highlighting the benefits and complexities of their implementation. Deng et al. [94] presented the concept of a DT city to address challenges in smart city development, discussing the necessary technologies and application scenarios.

Further advancements in DTs have focused on innovative frameworks and methodologies that integrate emerging technologies for improved disaster response and community resilience. Fan et al. [95] proposed an integrated textual–visual–geo framework to enhance social sensing in smart city DTs for disaster response, employing social media data for improved situational awareness and response capabilities. Ford and Wolf [96] discussed the necessity of integrating sensing and simulation across infrastructure systems for effective community management in smart cities, highlighting key issues in developing and deploying DTs for disaster management. Ham and Kim [97] introduced a framework for updating DT cities using crowdsourced visual data, enhancing risk-informed decision-making by providing real-time spatiotemporal information. Sun and Li [98] emphasized the importance of incorporating IoT (Internet of Things) and machine learning in DTs to advance urban emergency response planning, advocating for a participatory approach to urban resilience and sustainability. White et al. [99] demonstrated the use of DT smart cities for urban planning transparency and public feedback, showcasing a public DT of Dublin's Docklands with flood evacuation planning. Some other real-life works [100] leveraged city-scale DTs, exemplified by models from Helsinki and Lisbon, to enhance urban flood resilience through detailed 3D reality modeling, the integration of hydrometeorological and GIS data, and real-time flood simulation for effective risk assessment and emergency response.

The studies on DTs in the community resilience domain exhibit several limitations. Firstly, most of these studies are primarily conceptual, focusing on DT frameworks and models of communities rather than demonstrating the DT functions. Secondly, they tend to concentrate on single dimensions or specific components of the DT in development. This narrow focus leads to a fragmented understanding of communities, as it does not account for communities' intricate, multifaceted, interconnected, and dynamic characteristics, which is crucial in a complete understanding of community resilience. Lastly, the DT visualization aspects are overlooked, resulting in interfaces that lack user-friendliness, which is critical for decision-makers, the public, and stakeholder engagement. The visualization techniques used in DTs are also unintuitive and static [101], providing no dynamic data movement or ability for interaction. Adding simulations could make the visualizations more realistic and practical for users. These limitations hinder the true applications of DTs in practice and limit their potential in the community resilience domain.

This paper addresses these limitations by introducing COWIN^{E}, an ecosystem for community resilience strategies centered on DT technology. COWIN^{E} utilizes Cities: Skylines (C:S) as its base simulation engine integrated with real-world data to develop the community DT. The primary goal of COWIN^{E} is to instill disaster resilience in communities through active, collaborative engagement in its use across the general public, decision-makers, and other involved stakeholders in the event of disasters. COWIN^{E} offers a practical and

comprehensive digital representation of communities capable of capturing communities' dynamic, intricate, multidimensional, and interconnected structures. Equipped with user-friendly and intuitive interfaces, COWIN^{E} features dynamic, interactive DT visualizations. The following sections further detail COWIN^{E}, but first, this study's objectives and scope are presented.

3. Objectives and Scope

The objective of this study is fourfold: (1) to present COWIN^{E}; (2) to detail its development process; (3) to demonstrate the use-case application of COWIN^{E} in the event of a disaster through demonstrative, simulation-based case studies on the pilot region; and (4) to explore potential research directions using COWIN^{E}.

To achieve these objectives, in Section 4, the paper first introduces the details of COWIN^{E}, such as its various interactive components in a DT and the real world. Section 5 provides a detailed explanation of COWIN^{E}'s DT development, focusing on the pilot region and the comprehensive process of collecting and integrating real-world data from the area. Section 6 demonstrates a case study application of collaborative resilience planning in the event of a tornado, showcasing the use of COWIN^{E}. Section 7 explores the research directions for using COWIN^{E} in community disaster resilience contexts and also discusses the limitations of the study herein. Finally, the paper concludes with a summary and key highlights.

4. Community Twin Ecosystem (COWIN^{E})

COWIN^{E} is proposed and designed as an ecosystem that brings the DTs of communities and their users together, including the general public, decision-makers, and stakeholders, such as first responders, utility providers, engineers, nonprofit organizations, and local community leaders. The goal is to instill disaster resilience in communities in the event of disasters through active collaboration in its utilization. This collaboration is facilitated through the communicative use of COWIN^{E} before, during, and after a natural disaster, ensuring that disruptions are minimized, daily activities are maintained as much as possible, and recovery is swift and efficient. Figure 3 illustrates the schematic structure of COWIN^{E}, depicting how the various components of the DT and the real-world counterpart interact within the ecosystem.

COWIN^{E} consists of interacting elements of community DT and its real-world counterpart. In parallel to the five-dimensional DT [69,70], as discussed earlier, COWIN^{E}'s DT contains real-world data (*DD*) and the base simulation engine (C:S). The base simulation engine includes Services (*Ss*), Virtual Entity (*VE*), and Connections ($CN_{Ss\text{-}DD/VE\text{-}DD/Ss\text{-}VE}$) connecting the DT dimensions. The other part of Connections ($CN_{PE\text{-}DD/PE\text{-}VE/PE\text{-}Ss}$) connects the Physical Entity (*PE*—which is the real-world community) to the DT itself.

The collaborative aspect of COWIN^{E} comes into play through its user-friendly and intuitive online interface, which offers dynamic and interactive visualizations of the community DT. This aspect is very critical to making informed decisions rapidly during disasters. In an ideal collaborative use scenario, decision-makers and stakeholders utilize the operational observed data from the DT of the community in the event of a disaster to monitor the resilience performance of the community. These data then inform the critical analysis, evaluation, and interpretation processes, leading to decisions on which resilience practices to implement. The action items are then executed by the users within DT using its online interface.

If the action items meet the community resilience demands in DT and the community shows favorable resilience outcomes, they are formalized into a plan (action items in Figure 3) to be applied to the real-world community, which is the Physical Entity as depicted in Figure 3. This plan is executed in the real-world community with active collaboration with the general public to ensure their needs in the disaster event are understood and well-addressed. Following the implementation of these action items in the real-world community, observed data (Figure 3) from both the community DT and the real-world

community is continuously monitored iteratively to identify emerging vulnerabilities and opportunities for further enhancing the community's resilience.

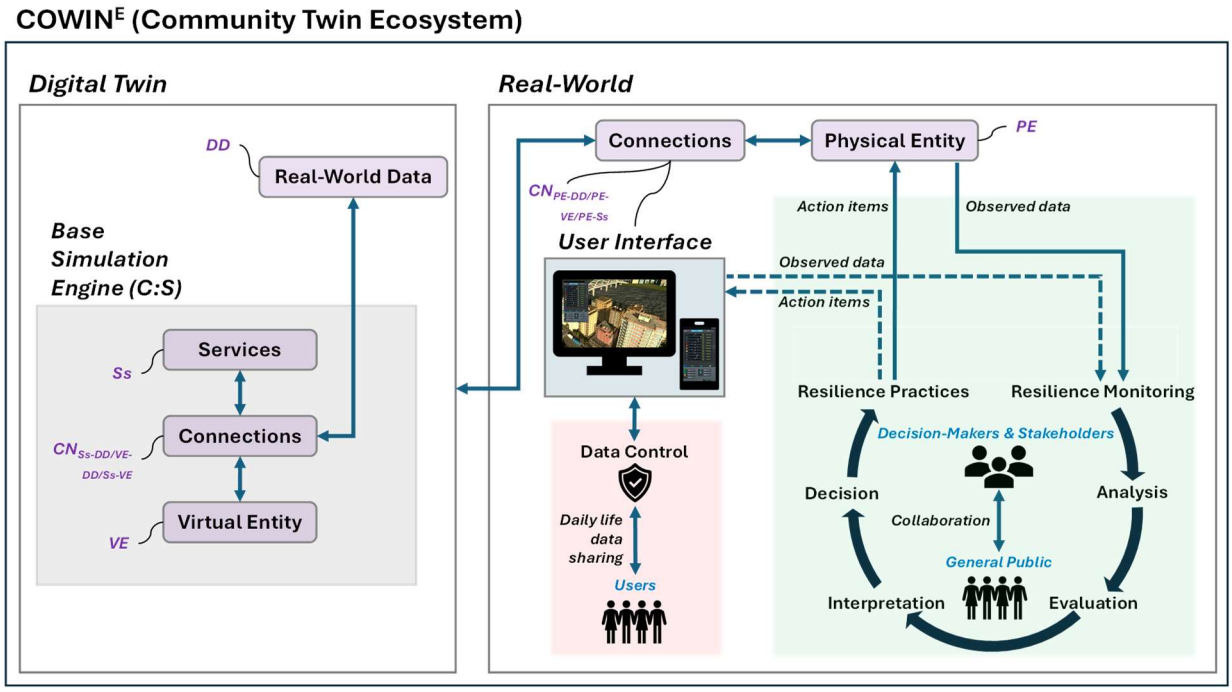


Figure 3. Schematic structure of COWIN^E. See Figures 2 and 4 for additional information about five-dimensional DT.

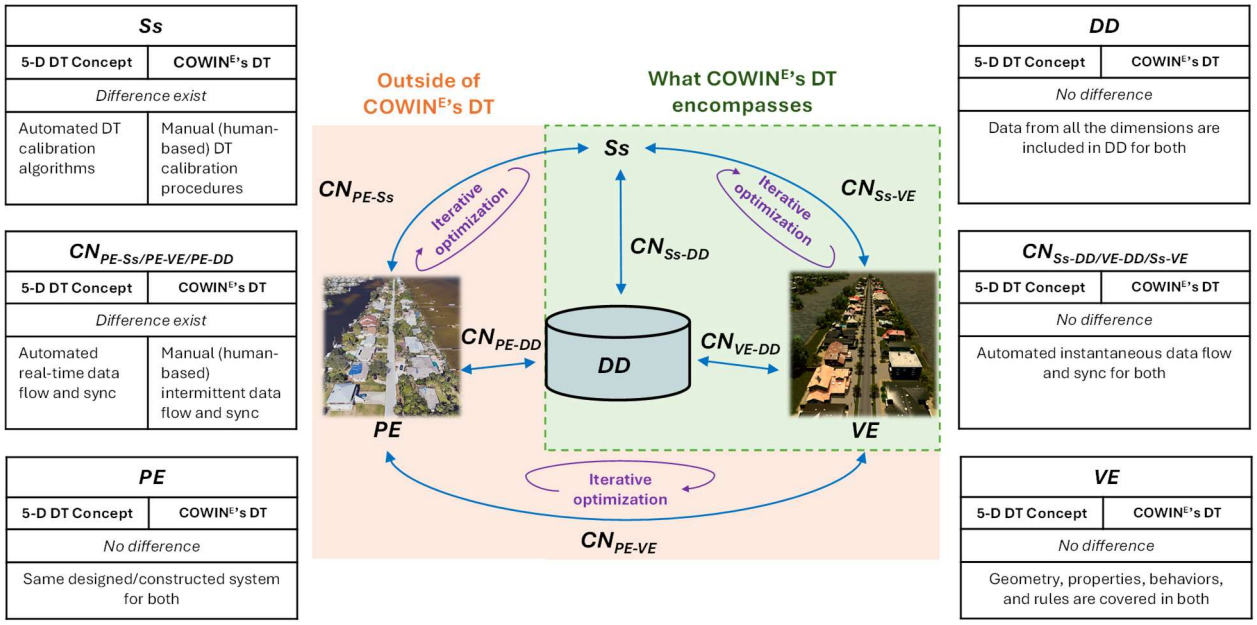


Figure 4. Position of COWINE's DT in the five-dimensional DT concept.

It is important to note that incorporating public participation—by integrating their knowledge, preferences, and feedback—strengthens the relevance and effectiveness of the resilience measures. The general public in COWIN^E is authorized to share vital data (with limited edit access, controlled by an additional security layer) related to their daily lives, such as the condition of their homes, local roads, or their daily needs. A case application

of such collaboration on resilience planning before, during, and after a tornado disaster through the use of COWIN^{E} is detailed in Section 6.

4.1. COWINE's Digital Twin

It is pertinent to explain how COWIN^{E}'s DT relates to the five-dimensional DT concept with a contrasting approach, as illustrated in Figure 4. The figure tabulates the critical differences between each dimension, linking to the five-dimensional DT.

The DT in COWIN^{E} inherently encompasses the *Ss*, *DD*, and *VE* dimensions and the dimension of connections among them, $CN_{Ss\text{-}DD}$, $CN_{VE\text{-}DD}$, and $CN_{Ss\text{-}VE}$. This encompassing means that COWIN^{E}'s DT internally handles the operating and processing of these dimensions within the DT base simulation engine, C:S. Consequently, $CN_{PE\text{-}Ss}$, $CN_{PE\text{-}VE}$, and $CN_{PE\text{-}DD}$, which facilitate communication between *PE* and other dimensions, are not part of COWIN^{E}'s DT. This is because the DT in COWIN^{E} does not incorporate automated real-time data flow and synchronization, which are the primary features of the connection dimension in the five-dimensional DT concept. There is, however, a human-based (manual) intermittent data flow and synchronization, where the human user takes the data once they are available from the *PE* dimension and utilizes them to update DT. The rest of the connections, $C_{NSs\text{-}DD}$, $CN_{VE\text{-}DD}$, and $CN_{Ss\text{-}VE}$, handle this data flow and synchronization instantaneously, as they are involved within the DT base engine. It is important to note that the term "instantaneous" is used for these connection dimensions because they enable an immediate and simultaneous response and update as they are within the DT base engine. Conversely, the term "real-time" is applied to the connection dimensions $CN_{PE\text{-}Ss}$, $CN_{PE\text{-}VE}$, and $CN_{PE\text{-}DD}$, as the data flow and synchronization from *PE* to other dimensions can experience delays in real-world cases due to different data transmission mediums (e.g., data transmission from physical sensors through cloud systems to DT).

For the *Ss* dimension, the means of DT calibration approaches differ. Manual or human-based DT calibration procedures are used to calibrate *VE* to align with the characteristics of *PE*. For example, once real-world data become available, the human user manually updates *VE* by adjusting its various aspects in DT as needed. In this detail, the iterative optimization process also falls into the human-based component to optimize the dimensions. However, the ideal method for this would be automated calibration algorithms for an effective and efficient DT environment.

Comparing COWIN^{E}'s DT to the five-dimensional DT concept does not yield more differences for other dimensions. For *PE*, the examined designed/constructed system would be the same for both. For *DD*, COWIN^{E}'s DT includes all the data *Dp*, *Dv*, *Ds*, *Dk*, and *Df*, as defined in the conceptual five-dimensional DT. For $CN_{Ss\text{-}DD}$, $CN_{VE\text{-}DD}$, and $CN_{Ss\text{-}VE}$, the automated, instantaneous data flow and synchronization exist for both COWINE's DT and the five-dimensional DT concept, as explained earlier. Similarly, for *VE*, the DT in COWINE includes the *Gv*, *Pv*, *Bv*, and *Rv*.

4.2. Base Simulation Engine

COWIN^{E} employs C:S as its base simulation engine for its DT development, utilizing C:S's advanced city simulation capabilities that allow users to plan, build, and manage urban areas, including zoning, infrastructure, transportation networks, and public services. The base simulation engine in DTs is the core computational platform that models, simulates, and replicates the behavior and dynamics of a physical system in a virtual environment, integrating data for predictive analysis and decision-making.

Although C:S is fundamentally a sophisticated city simulation 'game' engine widely used for entertainment purposes, it has been employed for various research goals due to its significant depth and realistic representation of urban dynamics. Some portion of these works studied the use of C:S for educational and training purposes in city planning, sustainability, and disasters [102–106]. Some other studies worked on the feasibility of using C:S in real-life urban planning, concluding highly promising evaluations of its application [101,107,108]. Increasing public participation in urban planning through the use of C:S

has also garnered significant interest in research [107,109–111]. C:S's interactive simulation capabilities allow citizens to engage directly with urban planning processes, offering a practical means to visualize and test planning proposals, thus fostering greater community involvement and feedback. C:S has also been utilized to address social and environmental problems in cities [112], urban sustainability, hazard, and epidemic management [113], and urban production [114].

C:S has been recognized in numerous studies for its highly detailed and realistic representations of various city aspects [101,104,106,107,109–111,113,115,116], highlighting its significant potential to be utilized as a city DT with a high degree of accuracy [105,112,113]. Thus, C:S forms the base simulation engine of COWIN^{E}'s DT. It is important to highlight some of the features and details of C:S, as presented in the following paragraphs. More detailed background about C:S is available in this reference [117]. C:S employs agent-based modeling, where autonomous units like citizens and vehicles interact within a complex urban simulation governed by rules that mirror real-life dynamics. Using the A* algorithm for pathfinding [118], C:S optimizes routes for vehicles and pedestrians, while procedural generation techniques create realistic cityscapes. C:S also features responsive traffic simulation algorithms that adapt to player modifications, along with physical, economic, environmental, and social simulations that reflect real-world challenges, which are programmed in C# and powered by Unity. This integration of diverse systems allows users to explore the impact of city planning, public services, and policies on urban development and citizen well-being.

Open Accessibility. One important feature that makes C:S unique and adaptable is that it allows developers to create content (modifications—shortly mods) and deploy it into the game, as well as to share it on the Steam Community platform so that other users can use these mods in their cities [119]. On this matter, C:S provides robust API and development tools that enable modders (mod creators) to introduce new features, assets, and gameplay mechanics [120]. This has led to a vibrant community of modders who have expanded the C:S's capabilities, possibly making city representations even closer to real-life scenarios. For instance, a study by Pinos et al. [121] developed a mod that automates the creation of geographically accurate visualizations of real-world places within the game. COWIN^{E} employs some of the publicly available mods, which are available in this reference [122].

5. Developing DT for the Pilot Region

5.1. Pilot Region in Brevard County

The COWIN^{E}'s DT is developed considering the broader area of Merritt Island and Cocoa in Brevard County, Florida, including Rockledge and Sharpes, as shown in Figure 5.

The primary reason for selecting this specific region in Florida's Brevard County for DT development is its significant strategic importance, connecting critical industries to the rest of mainland Florida. These industries include the public and private aerospace sectors, including NASA's Kennedy Space Center, SpaceX, Blue Origin, launch sites, and rocket factories; the defense sector, particularly the Air Force; and the tourism sector, such as Port Canaveral, which is the second busiest cruise terminal in the world, and the iconic Cocoa Beach, among other renowned beaches. Secondly, due to the region's archipelago-like tropical topography, the infrastructures, including utility systems and long multi-span key bridges, cross over lakes and ocean inlets, making them critical to the region's functionality. Lastly, Brevard County is significantly prone to frequent natural disasters like other parts of Florida, such as hurricanes, tornadoes, lightning, floods, wildfires, and heat waves, making these first two reasons immensely important. According to the National Risk Index community report for Brevard County [123], the county is highly susceptible to natural hazards such as hurricanes. Its relatively high-risk index score of 99.4 out of 100 places it in the 99th percentile nationally, meaning only less than 1% of U.S. counties have a higher risk. In Florida, Brevard County is in the 93rd percentile, indicating a higher risk than 93% of Florida counties, making the pilot region among the most at-risk areas in the US.

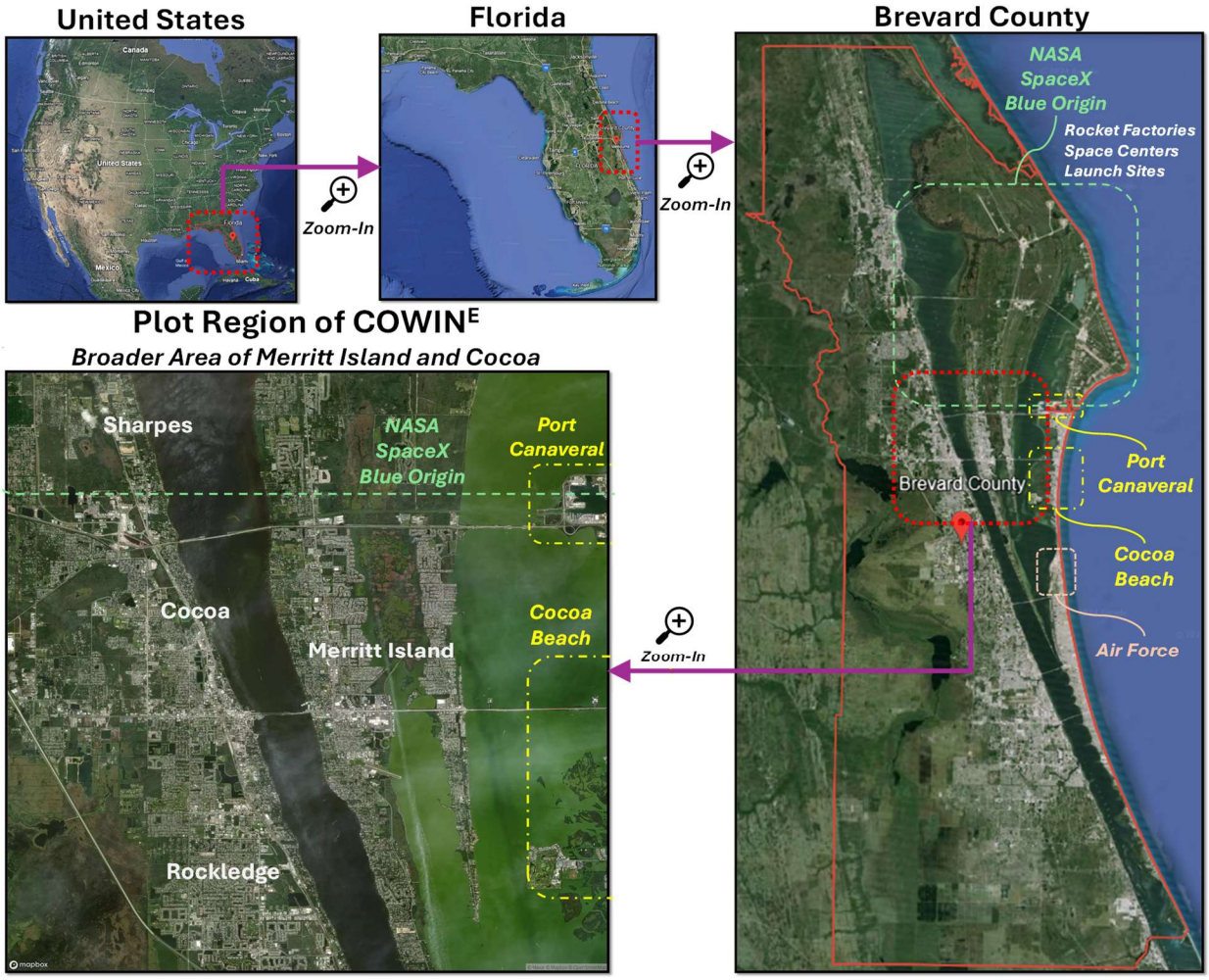


Figure 5. COWINE's pilot region in Brevard County, Florida: Broader area of Merritt Island and Cocoa.

In the event of one of these disasters or a cascading or co-occurring multi-disaster system, these critical industries, key infrastructure systems, and residential and commercial areas, along with socio-economically challenged communities, may suffer irrecoverable damages. On this matter, the county's composite EAL (Expected Annual Loss) is approximately USD 414.38 million, primarily driven by hurricanes and wildfires [123]. Specifically, hurricane EAL is the highest at USD 365.97 million, followed by wildfire at USD 18.89 million, and tornado at USD 11.07 million. Brevard County has experienced multiple hurricanes over the past decade, including ones like Matthew (2016), Irma (2017), and Ian (2022). Nationally, Brevard's EAL is in the 96.12th percentile, meaning it experiences higher annual losses from natural hazards than 96.12% of U.S. counties. In Florida, Brevard County ranks in the 85.83rd percentile, indicating it has higher expected losses than 85.83% of counties in the state. This highlights Brevard County's significant vulnerability to natural hazards, leading to considerable anticipated economic losses.

While Brevard County faces significant risks from hurricanes, wildfires, and tornadoes, with high EALs, a report [123] indicates that the county has a moderate social vulnerability score of 47.1 and a resilience score of 59.8 (out of 100), which indicates a balanced susceptibility to the adverse impacts of natural hazards and moderate community resilience. This shows that, despite the high risk and potential consequences of the disasters, the community's vulnerability to natural hazards is moderate, and its resilience is also moderate. These scores suggest that the county's current level of preparedness and resilience is likely not sufficient to manage and mitigate the impact of such disasters effectively. Thus,

this region is a suitable candidate as a case-study region in COWIN^{E} to show its disaster resilience performance.

5.2. Real-World Data Integration

As the DT is a representation of the broader area of Merritt Island and Cocoa, it integrates various real-world data sources with varying levels of detail, depending on their availability in public databases. These data sources include the region's topography, infrastructure, residential, commercial, industrial, and office areas, economic structures, city policies, natural disaster statistics, traffic information, and public transit systems, which are illustrated in Figure 6.

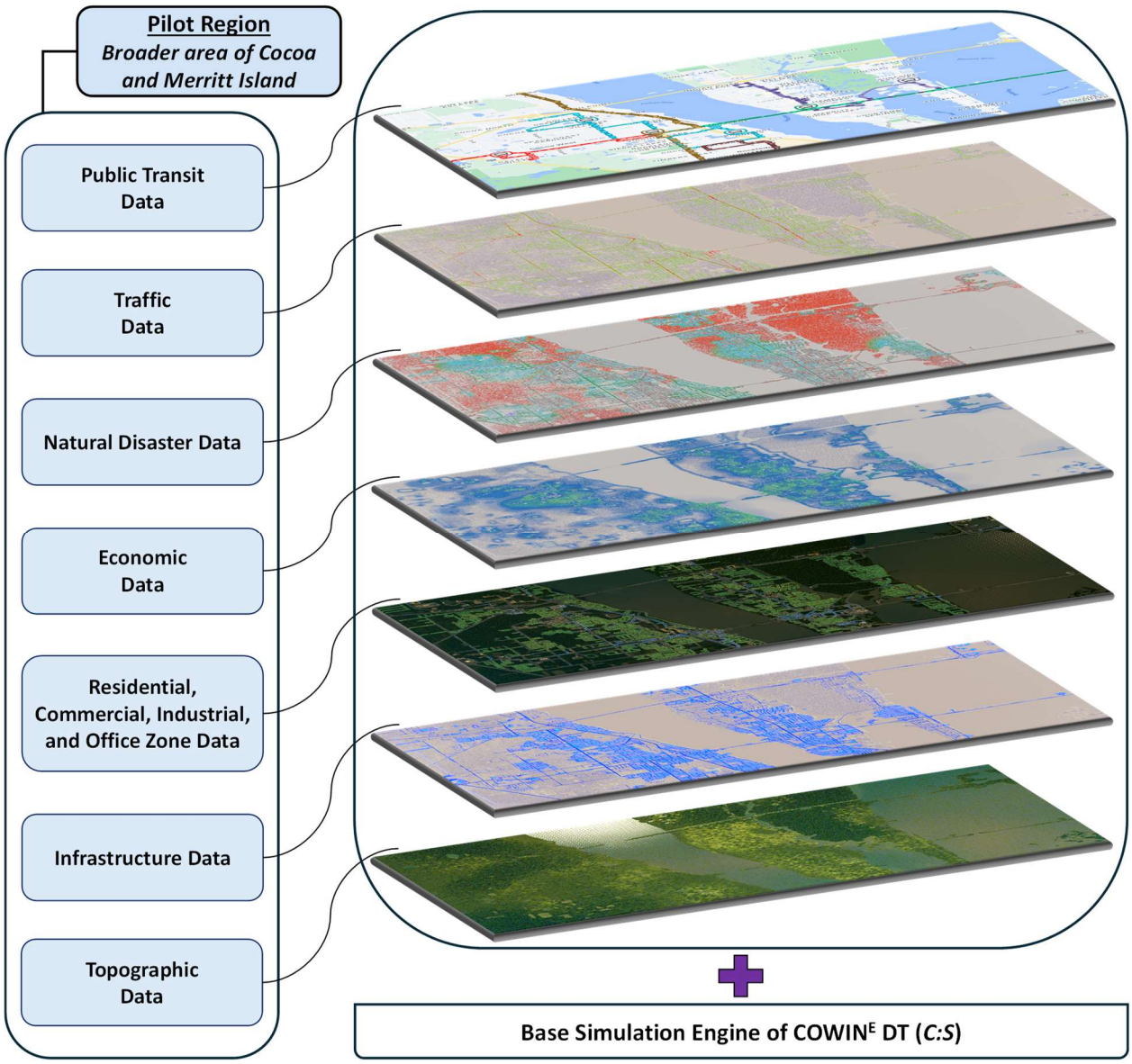


Figure 6. Data sources utilized in developing the DT.

These data sources are integrated into the COWIN^{E}'s DT base simulation engine through C:S's built-in features, such as its embedded physical assets, predefined mechanisms, or other user interface components. These features allow building different topographic maps (e.g., generating lakes, hills, oceans), infrastructure systems (e.g., building power plants, water treatment systems, schools, fire stations), setting up growable resi-

dential, commercial, industrial, and office areas (which develop into physical structures as the city develops during the simulation), defining economic parameters and policies (e.g., tax rates, budget allocations, incentives, and regulations), implementing traffic and public transit systems (e.g., road networks, bus lines, metro systems), and simulating the effects of natural disasters (e.g., hurricanes, floods) on the region's infrastructure and communities.

However, modeling and developing real-world urban environments and their intricate dynamics in a digital environment can be a significantly complex task. Although C:S offers these built-in features to develop and model cities, it is still challenging to match the real-world settings in a digital environment. Hence, throughout the integration of the real-world data, this paper also utilized numerous mods (modifications) [123] created by developers, as explained earlier, to compensate for what the base simulation engine lacks in terms of built-in features for matching real-world dynamism. Using these mods helps bring urban representation closer to real-world scenarios both in urban realism and functionality. For instance, one of these mods, TM:PE (Traffic Manager: President Edition), aids in making more detailed traffic management that mirrors real-world conditions, such as controlling traffic signals, setting lane priorities, managing speed limits, and various other traffic rules. These features enable a more accurate simulation of urban traffic dynamics. One other mode is Real Time, which makes city life more realistic by aligning citizens' activities with real-world schedules (e.g., rush-hour times, lunch hours, or school start times—which can be externally modifiable), leading to natural fluctuations in traffic and demand for services throughout the day. In addition, real-world assets can also be integrated into the urban environment. As such, this paper utilized some of the commonly available commercial locations in Brevard County, e.g., Walmart, McDonald's, Subway, Hyundai, and Publix, and some of them are illustrated in Figure 7.

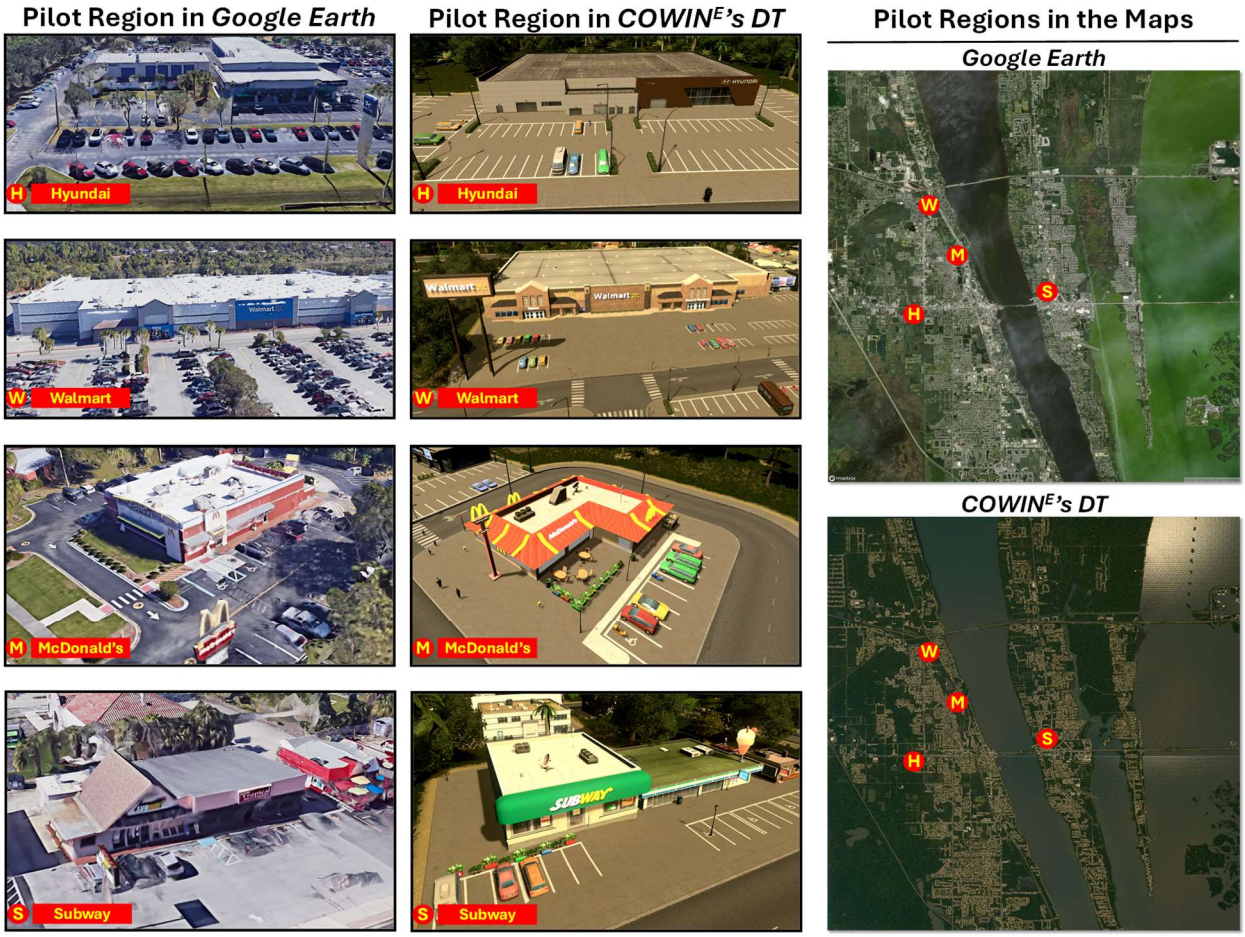


Figure 7. Some of the real-world commercial places included in DT.

5.2.1. Topographic Data

Prior to integrating the infrastructure data, the region's topographic map is incorporated, as illustrated in Figure 8. To do so, the heightmap of the region (which provides terrain elevations for a broader area of Merritt Island and Cocoa) is obtained from the Heightmap Generator website [124]. Followingly, the target square on the website is positioned in the selected region. Then, the map size is set to 17.28 by 17.28 km, which aligns with a one-to-one real-scale representation. Although the full 81-tile grid in the base simulation engine measures 18 by 18 km, this slight adjustment ensures more accurate scaling. Subsequently, the heightmap image of the region is downloaded from the website as a PNG file (using the buttons in the left toolbar) and then placed in the C:S directory. Finally, the heightmap is imported through the C:S map editor's "Import Heightmap" feature, bringing the terrain data into the base simulation engine.

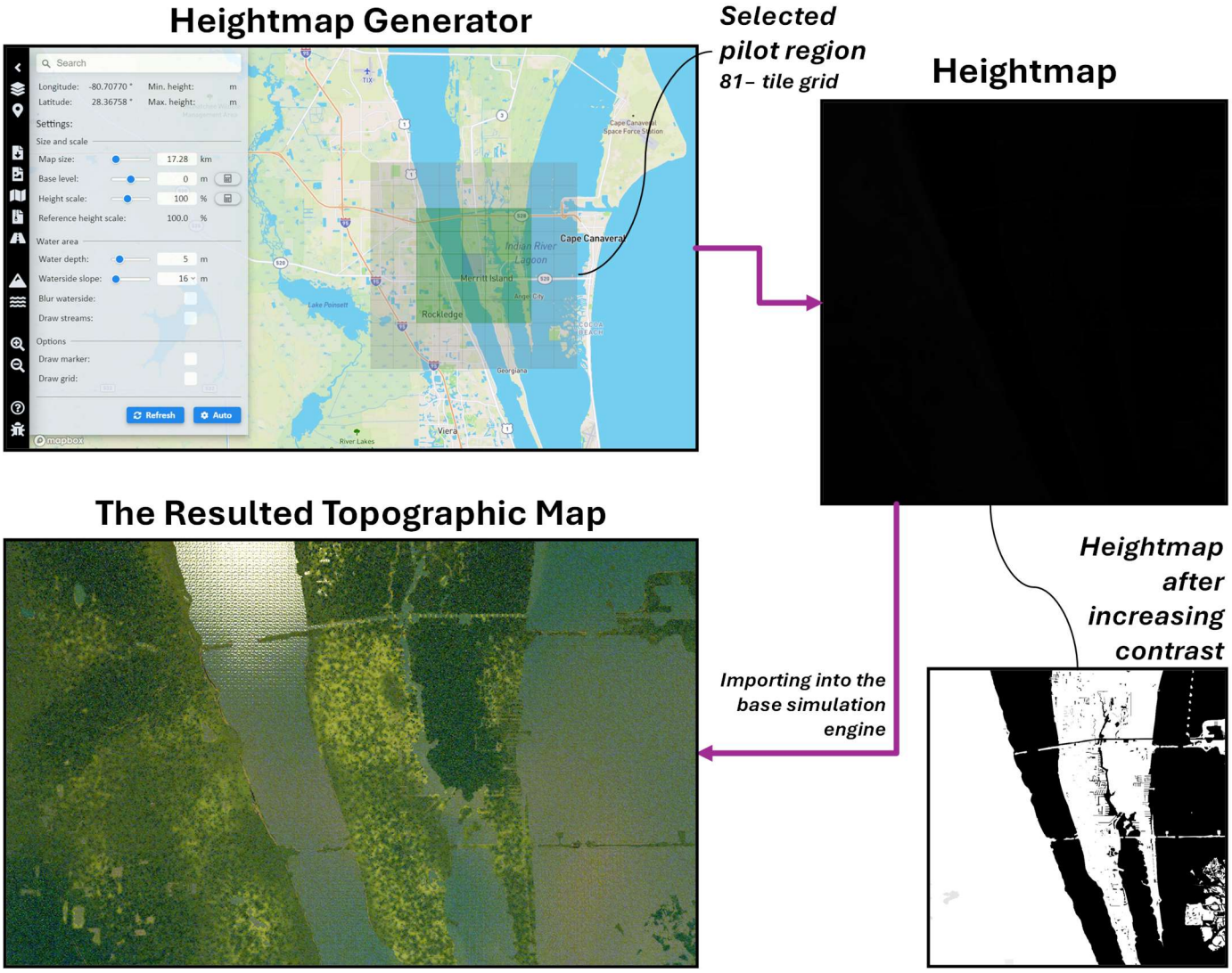


Figure 8. Process of importing the topographic map of the pilot region into the base simulation engine.

The discrepancy between the 18 × 18 km and 17.28 × 17.28 km map sizes arises from how C:S handles scaling. While each tile in the full 81-tile grid measures 2 × 2 km, resulting in an 18 × 18 km area, this size slightly exceeds the actual area in C:S when aiming for precise one-to-one real-world scaling. Hence, using a 17.28 × 17.28 km map ensures that the in-simulation representation of real-world locations is more accurate, compensating for the way C:S renders terrain and distances.

Note that the heightmap image of the region is mainly black, showing that the elevation data in that region are uniform and very low, which can happen with flat regions like coastal areas or plains, like Florida. Heightmap images use grayscale to represent elevation, with black representing the lowest points (often sea level) and white representing the highest elevations. If the entire area is at a similar low elevation, the heightmap may appear as a solid black image. Following the contrast adjustment in the image, terrain elevations can be seen in Figure 8 (but still, there are several ponds and lakes in the area that are not visible in the contrasted image, which can be visualized with more contrast).

Lastly, after importing the topographic data, the water stream (a built-in feature of C:S) is applied to the ponds, lakes, and ocean inlets, and trees and vegetation commonly found in Florida are placed in the map, which are palm, pine, and oak trees. These are implemented using the Image Overlay Renewal (IOR) mod, which is explained in the following section.

5.2.2. Infrastructure Data

This paper considers infrastructure systems such as power and water plants, networks, road and bridge systems, and other facilities (e.g., healthcare, emergency, law enforcement, education, parks, recreation, and waste—excluding the residential, commercial, industrial, and office areas) that serve the daily needs of the public. To integrate the data from these infrastructure systems into the base simulation engine, the mod IOR is leveraged. This mod allows users to import and overlay custom images, such as satellite maps or blueprints, in the background of the map, which helps in the alignment of real-world locations in the base simulation engine for building the infrastructure elements, just like drawing on a tracing paper for overlay purposes.

This paper utilizes two different images of the region, street and satellite images, to overlay the real-world images in the background of the topographic map (the topographic map was imported into the base simulation engine in the previous section). These images, like the heightmap image obtained from the Heightmap Generator, are downloaded as PNG files using the left toolbar. They are then placed in the C:S directory. The images are then visualized in the background of the topographic map by adjusting the image opacities via the IOR's hotkeys. This helps align the positions of the region's infrastructure elements when manually placing different infrastructure systems in the base simulation engine.

It is important to note that, at this stage of the research, the capacities of these infrastructures (e.g., hospital beds or fire station truck capacities) are generally not considered as they are not included in this paper's scope. This will be taken into account in the next study as they will be critical factors for future research.

Roads and Bridges. The IOR mod is especially useful for building roads (including railroads and cargo train terminals) and bridges in the base simulation engine, as it allows the roads and bridges to be visible in the background during their building process. While placing roads and bridges based on these two images overlaid, additional satellite and street 3D views from Google Earth are also referenced in another monitor and iteratively updated in the base simulation engine (e.g., checking the number of lanes and junctions). Discrepancies in road alignment are often observed when comparing the overlaid images (sourced from Mapbox and OpenStreetMap) to Google Earth maps, showing that Google Earth provides more up-to-date views. The illustration in Figure 9 shows the IOR mod in both its "ON" (top left) and "OFF" (top right) states. The figure also shows the views of the Route 528 bridge, as seen in Google Earth and in COWINE's DT (after the roads and bridges are included). The bridge is critical for the transport of goods and people during regular times as well as extreme event situations.

Electrical and Water Systems. The data for power plants, electrical substations, and transmission lines for the pilot region were sourced from OpenInfraMap [125]. The region's power is supplied by Plant Oleander, a natural gas plant with a backup fuel oil option, which is distributed to the region through transmission lines and several substations. These power plants, transmission lines, and substations are built into the base simulation engine

using C:S's built-in assets of natural gas and fuel oil plants, along with transmission lines and substations (added via mods) to make the energy supply a total of 994 MW. For the water systems, due to the unavailability of open-source data, the details of the water pumping stations, sewage and freshwater outlets, and water pipelines are assumed in the pilot region. On the other hand, the location data for the water treatment plants and water tanks were obtained from Google Earth. These water systems are then included in the base simulation engine using C:S's built-in water-related assets.

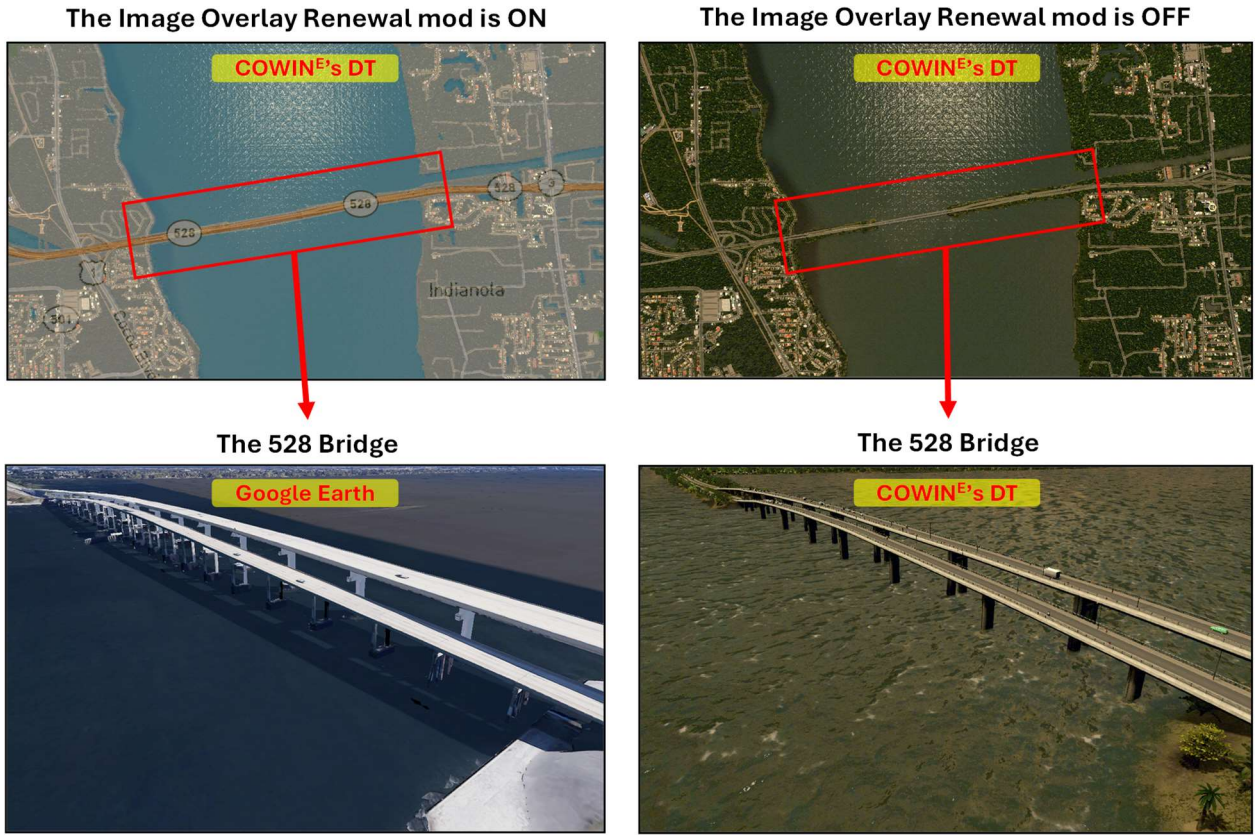


Figure 9. Some example illustrations of the Image Overlay Renewal mod "ON" (**top left**) vs. "OFF" (**top right**) and views of the Route 528 bridge in Google Earth and COWIN^{E}'s DT.

Healthcare, Emergency, and Law Enforcement. The location details of healthcare, emergency, and law enforcement in the pilot region are readily available online. This paper utilized ArcGIS data [126] and Google Earth maps extensively to obtain these data. As a result, several healthcare, emergency, and law enforcement units are built into the base simulation engine using the built-in assets. These involve hospitals, medical clinics, elder care, child health centers, medical helicopter depots, cemeteries, crematoriums, fire stations, fire watch towers, fire helicopter depots, disaster response units with airbase supports, shelters or resilience hubs, radio towers, weather and space radars, and police departments.

Education Facilities. Similar to healthcare, emergency, and law enforcement, placing education facilities in the base simulation engine process was completed by referencing ArcGIS data [126] and Google Earth maps, as all the related data are there. Consequently, several elementary schools, middle schools, high schools, universities, and libraries are placed in the base simulation engine using the built-in assets. They can also serve as hurricane shelters with additional mod extensions.

Parks and Recreational Facilities, and Others. ArcGIS data [126] and Google Earth maps were referenced to place the parks and recreational facilities in the base simulation engine using the built-in and mod-based assets, similar to other infrastructure data. These include various parks (parks like natural outdoor spaces for recreation and relaxation, special gar-

dens, and playgrounds), sports facilities (e.g., fitness centers, tennis and basketball courts, football fields, and stadiums), museums, hotels, supermarkets, and shopping centers.

Government and Religious Buildings. As for the government and religious buildings, similarly, ArcGIS data [126] and Google Earth maps were referenced. These buildings involve the placing of various churches, county courts, tax offices, city halls, and other government offices in the base simulation engine via the built-in and mod-based assets.

Waste Facilities. Waste facilities include landfill sites and recycling facilities. These are placed in the base simulation engine, referencing ArcGIS data [126] and Google Earth maps via the built-in assets.

These infrastructure systems, as presented in the earlier paragraphs, interact with community dynamics in different means through C:S's various built-in mechanisms. For instance, electrical and water systems directly impact population satisfaction, city growth, and overall functionality, as they are essential for providing basic utilities. Roads and bridges provide access to services, improve land value, and foster population growth and satisfaction by ensuring efficient transportation and connectivity. Healthcare, emergency, and law enforcement maintain public order and stability in the city, which ensures the well-being and safety of the population, reduces crime rates, and responds to disasters. Education facilities improve population education levels, leading to a more skilled workforce, increased land value, and overall higher city development and productivity. Parks and recreational facilities impact the happiness level of the population, which affects the overall satisfaction and well-being of the population, encouraging higher residential, economic growth, and stability. Government and religious buildings enhance land value, boost population happiness, and increase the attractiveness and cultural significance of the city while being critical for operations and management. Waste facilities manage garbage and sewage, crucially preventing pollution and maintaining population health, which directly impacts overall city happiness and land value.

5.2.3. Residential, Commercial, Industrial, and Office Data

Although some of the notable commercial retailers were placed in the base simulation engine earlier (Figure 7), the commercial, residential, and industrial zones are included separately. This is because the built-in tool of 'zoning' in the base simulation engine functions differently than placing the individual assets in the maps, such as placing the fire stations or hospitals. Zoning involves designating specific areas that correspond to real-world locations, whether residential, commercial, industrial, or office, rather than simply placing pre-built assets. These zones then grow and evolve as the city develops.

The data for residential, commercial, industrial, and office zones were similarly obtained from ArcGIS [126] along with Google Earth maps for additional verification purposes. Placing these zones follows the same strategy as the one implemented for integrating the infrastructure data, which leverages the mod IOR. It is noteworthy that C:S offers specific industrial assets, rather than zones, which include elements like wood plants and log yards for the forestry industry, animal pastures, and barns for the farming industry, as well as warehouses and postal service facilities. These assets are also integrated into the base simulation engine to correspond with their real-world counterparts.

5.2.4. Economic Data

Representing economic dimensions of real-world systems can be a very complex task, particularly for large systems like cities. C:S provides a simplified yet effective economic mechanism based on the income and expenses from different city components (e.g., income from tourism, jobs, industries, bus tickets, manufacturing, taxes, etc.). It also offers an interface for controlling taxes for different zones, as well as managing budgets for various city aspects (e.g., education, emergency services, and utilities).

At this stage of the research, this paper defines the tax rate for each of these zones—residential, commercial, industrial, and office—as 7%, which is the sale tax in Brevard County. It is acknowledged that every individual or corporation in these zones is taxed

based on various factors, such as property assessments, exemptions, and local regulations. Actual tax rates can vary due to the influence of local tax policies, the assessed value of the property, and the characteristics of each zone, which contribute to a more complex and dynamic tax environment that would need to be addressed in further stages of this research for a more accurate representation. City budget and loan data are also not included in the base simulation engine, which may be considered in future research.

5.2.5. Natural Disaster Data

The most common disasters in the pilot region are hurricanes, tornadoes, and wildfires, as indicated earlier. The base simulation engine of COWIN^{E} DT uses the C:S's downloadable content of natural disasters, which allows users to simulate various types of disasters in cities and to manage evacuation routes and different emergency services to manage the city disaster preparation. The disaster amplitudes in C:S can be adjusted using a scale from 1 to 10, allowing the control of the severity of each disaster type. However, the underlying basis for this scale is not clearly defined by C:S. Refining it with a mod to align with realistic disaster magnitudes for each disaster type can enhance the game's realism.

Integrating the real-world disaster data into the base simulation engine is carried out by leveraging a Natural Disasters Overhaul mod. This mod implements the disasters based on their probability per year values and peak seasons. Only the disaster data of storms (hurricanes, tropical depressions, and storms), tornadoes, and wildfires are defined using this mod, though this could also be extended to other disasters. The probability of storms and tornadoes occurring in Brevard County is identified as 1.41 and 1.94 per year, respectively, with their peak seasons being September, based on historical data from NOAA [127]. For wildfires, it is 7.97 per year, identified in a different source [128].

5.2.6. Traffic Data

Integrating and modeling traffic data in the base simulation engine is a major task and has critical importance to the functionality of the city. While traffic data generally involve a combination of dynamic numbers, such as vehicle volumes, their speeds, traffic incidents, and congestion levels, these are typically the outcomes of established traffic rules, road designs, population, and dynamics of the zones connected (e.g., residential or office zones). C:S and its modding community do not offer a mechanism to adjust any of these dynamic traffic data for now. Thus, the base simulation engine of COWIN^{E}'s DT only considers the pilot region's traffic rules set for the roads, which defines the region's dynamic traffic or mobility data as the city grows.

This paper used various types of data available online for setting traffic rules, such as FDOT speed limits [129], number of lanes [130], traffic signal locations [131], and Google Earth maps, as a reference. Nevertheless, some other additional traffic data are still needed, such as turn restrictions, lane merges, pedestrian crossings, and traffic signs, and for that, Google Earth maps were used. These traffic rule-related data were then integrated into the base simulation engine utilizing the mod TM:PE described earlier.

5.2.7. Public Transit Data

Although the population in the U.S. generally relies more on personal vehicle usage, the pilot region has a well-scheduled public bus network with good coverage of the area. Modeling such a network is of utmost importance for accurately reflecting the transportation dynamics of the region and ensuring that the network meets the needs of the community. This paper used the public transit data available online [132].

Brevard County has 23 bus lines, and the pilot region includes 7 of these lines (e.g., a bus line that covers Cocoa and Rockledge or West Cocoa). To integrate these bus lines, including the bus stops, in the base simulation engine, C:S's interactive public transportation feature is used. This built-in feature allows the creation of bus lines and stops with the number and types of buses needed in the region, which C:S calculates and assigns automatically. The number and types of buses can also be adjusted manually. Based

on [132], two to three buses were assigned for each line, but this also changes dynamically for Brevard County.

6. Case Study: Collaborative Resilience Planning for a Tornado

Following the integration of real-world data into the base simulation engine, COWIN^{E}'s DT was run (simulated) until the population (agents) had completed settling in all the designated residential zones and the commercial, office, and industrial zones developed. This simulation process took roughly three months to complete and was implemented under human supervision to ensure accuracy in the data inputs and alignment with expected urban development patterns, ultimately providing a validated model of the city's potential growth and infrastructure dynamics. In the end, the population in the region reached 120,000 while the real population of the region, according to the official reports, is roughly 100,000, and this is anticipated to be higher when totaled with unofficial numbers. The analysis and comparison of the demographic numbers in the DT and real-world regions will be covered in the subsequent studies.

It is important to explain the core resilience properties [25,133] as they are referred to in the DT's collaborative use case. These properties, *vulnerability*, *robustness*, *redundancy*, *resourcefulness*, *rapidity*, and *adaptation*, are integral to resilience, as depicted in Figure 10. Vulnerability represents the system's initial susceptibility when exposed to the disaster, illustrated by the immediate dip in functionality. Robustness is the system's ability to withstand disturbances without significant loss of functionality, while redundancy involves having backup systems or excess capacity to maintain operations during crises, which is illustrated as a quick recovery jump in Figure 10, followed by resourcefulness with a little delay. As the disaster impacts the system, responsiveness 1 captures the period of immediate actions taken to stabilize the situation. Resourcefulness reflects the system's capacity to manage and deploy resources effectively and determines the overall success of the response and recovery efforts. Rapidity is demonstrated in the speed with which these actions are taken. The later period of the system's recovery is captured by responsiveness 2. Lastly, adaptation illustrates the system's ability not only to recover but also to evolve, improving resilience and functionality beyond pre-disaster levels.

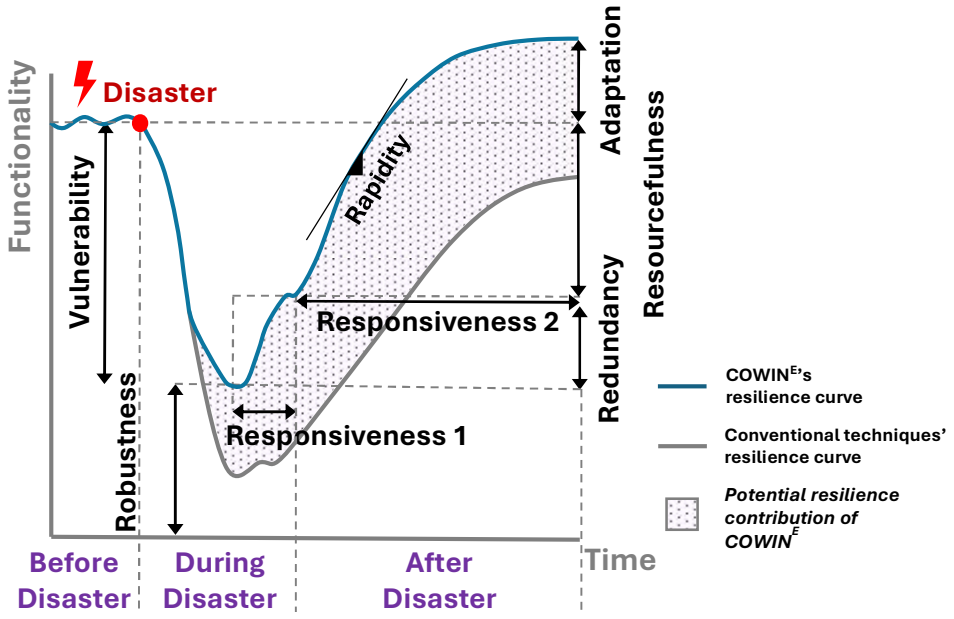


Figure 10. Hypothetical resilience curve illustrating the core resilience properties.

Referencing the resilience properties for the actions of stakeholders and the general public (e.g., how stakeholders' actions influence the resilience properties), the DT's use example case is illustrated before, during, and after the tornado impact in Figures 11–13.

Note that some of these actions may influence (affect) multiple resilience properties directly or indirectly, and only the most predominant ones are mentioned. In addition, rapidity is not directly referenced as it is the derivative or rate of change, essentially representing the slope of resourcefulness over time. For the sake of simplicity, only some of the key stakeholders engaged in the event of a disaster are included in this example, such as emergency response coordinators (decision-makers), meteorologists, first responders (firefighters and search and rescue teams), the general public (residents), utility companies, nonprofit organizations, structural engineers, community leaders, and meteorologists. These stakeholders use the DT on their phones/computers to access community dynamic data while seeing the active urban visuals and predictive simulations that guide their decision-making processes. Subsequently, they input their decision (or actions) directly into DT via their devices and message other stakeholders, which is denoted as "Updating in DT" in the figures. If private conversations are made between the stakeholders, then it is denoted as "Contacting…".

6.1. Before Disaster

As illustrated for the before tornado case in Figure 11, emergency response (ER) coordinators identify the high-risk areas with older infrastructure, like bridges and utility networks, based on the forecasted data in the DT (indicated as 'twin' in figures). They then disseminate the region's vulnerability information to relevant stakeholders using the DT to ensure that first responders, utility companies, and community leaders are all informed and can take appropriate pre-emptive actions. The circled number 1 attached to the ER coordinator dialogue indicates that ER coordinators' actions influence the vulnerability aspect of resilience, as reflected in the resilience curve in Figure 11. Structural engineers indicate the condition rating [134] of the critical bridge (Huber Humphrey) as poor, 4 out of 9, based on the NBI (National Bridge Inventory) condition rating, and the tornado simulation in DT predicts catastrophic bridge failure. Thus, the engineers suggest an alternative bridge route as the 528 bridge. This action influences the vulnerability (number 1) and redundancy (number 3), as the engineers provide information about the vulnerability of the bridge and offer another alternative to be used, which is a backup plan and affects redundancy.

First responders, on the other hand, relocate the excessive number of helicopters from Cocoa to Merritt Island for backup in the fire stations after the DT's indication of a potential strike in Merritt Island. This affects the redundancy and resourcefulness properties of resilience as an additional layer of operational capacity is created by using extra resources. Utility companies focus on securing power lines and substations based on the DT's predictive simulation, actions that bolster the robustness of the community (number 2). The DT identifies further vulnerabilities in the power grid (number 1) and updates this information in the DT. Nonprofit organizations are alerted to prepare backup shelters and stockpile essential supplies in anticipation of potential disruptions (numbers 3 and 4). Meanwhile, public users who are part of an underrepresented community in the region (as in this case) reach out to the community leaders for disaster preparedness, asking questions about evacuation orders and additional supplies, influencing resourcefulness (number 4).

6.2. During Disaster

During the disaster, the tornado makes landfall, and various stakeholders actively respond to the unfolding crisis using the DT, where the predictive simulations in the DT guide the decisions and actions of stakeholders and the general public. The during-disaster case is illustrated in Figure 12.

Meteorologists note the slight shift in the potential strike area with increased intensity in the DT. ER coordinators manage the overall response by issuing evacuation orders and monitoring the ground status updates through the DT, which influences resourcefulness (number 4). First responders report that the tornado strikes Cocoa and is currently moving toward Hubert Bridge while reporting several casualties. This information influences the robustness of the emergency response system (number 2) because it shows the strength

of the emergency response system in managing evolving situations under stress and continuing to deliver critical services. The DT predicts a potential deck failure on Hubert Humphrey Bridge as the high-intensity tornado approaches. Structural engineers also verify and concur with this predictive simulation. This is critical because, in the before-disaster phase, structural engineers indicated the bridge's poor condition and suggested the use of the 528 bridge as part of the evacuation route as per the DT's predictions, which influenced the vulnerability and redundancy in the before-disaster planning. Now, the prediction about the potential deck failure (after the update from the meteorologists) and structural engineers' confirmation influences the vulnerability of the region (number 1).

Utility companies observe the broken power lines near Cocoa and indicate that that power line will not transmit electricity to South Merritt Island, mentioning that they will send out a crew to repair the line. They also note in the DT that the power line by the 528 bridge remains operational and supports Merritt Island. Their actions influence resourcefulness (number 4) through the repair decision of the power line for immediate recovery and redundancy (number 3) through the indication of the other operational power line by the 528 bridge. As shelters begin to fill up, non-profit organizations respond by setting up additional shelters and updating the DT with the shelter's new capacity. Their proactive approach demonstrates redundancy and resourcefulness (numbers 3 and 4) as they ensure that there is adequate shelter space for evacuees, even as the disaster unfolds. Community leaders and the public maintain active communication through the DT, ensuring that residents are informed and able to make decisions. This interaction emphasizes redundancy and resourcefulness (numbers 3 and 4) as residents adapt to the evolving situation using the DT's updates while community leaders address concerns about shelter capacity and backup power.

6.3. After Disaster

After the disaster, as illustrated in Figure 13, stakeholders use the DT to guide the recovery efforts. ER coordinators manage the overall recovery process by prioritizing rescuing people, repairing the bridge and power lines, and assessing the damage to critical infrastructure. This influences resourcefulness (number 4) by ensuring efficient resource allocation and swift restoration of essential services. First responders continue their search and rescue operations on Florida Avenue using ground and aerial robots, rescuing dozens of people, which influence the robustness and resourcefulness of resilience (numbers 2 and 4). These actions influence robustness by ensuring the emergency response system can effectively operate under challenging conditions using advanced technology, and they affect resourcefulness by efficiently coordinating teams and maximizing the use of available resources via DT. Utility companies report that power lines on Merritt Island have been repaired, and power has been restored. They will also analyze the cause of the line failure and implement more robust power transmission line strategies based on the lessons learned. These actions reflect the importance of resourcefulness and adaptation (numbers 4 and 5), ensuring that the infrastructure is not only restored but will also be improved to withstand future disasters with better capacity.

Structural engineers, informed by the DT, are on the field sending robots into high-rise buildings for rapid assessment and heading to the collapsed bridge to draft a repair plan. Based on their findings, they will update repair and retrofit recommendations and plan to address newly identified vulnerabilities. Their actions influence resourcefulness and adaptation (numbers 4 and 5) by effectively utilizing technology resources to gather critical information and adapting building practices to improve resilience against future disasters. Nonprofit organizations coordinate with community leaders through the DT, distribute essential supplies, and provide temporary housing to affected residents. The DT's updates ensure that residents know where to go for assistance, showcasing resourcefulness and redundancy (numbers 3 and 4) in managing and distributing aid resources effectively and providing extra housing units. Community leaders maintain active communication with the public through the DT, providing guidance on safety, repair efforts, and when it will be

safe to return home. This interaction reinforces the importance of resourcefulness (number 4), ensuring that residents remain informed and can make decisions based on the most current information available.

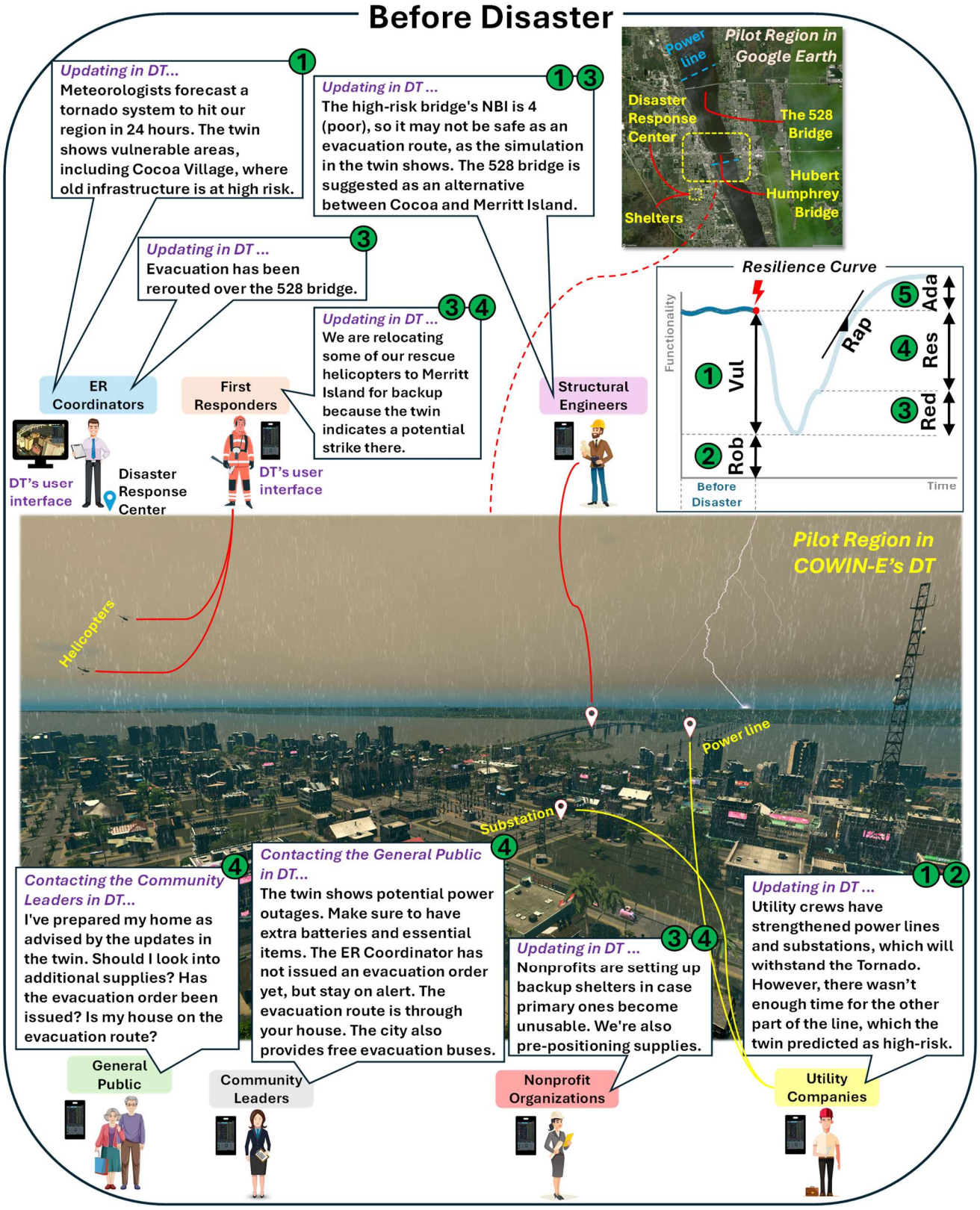


Figure 11. Collaborative use of DT before the tornado.

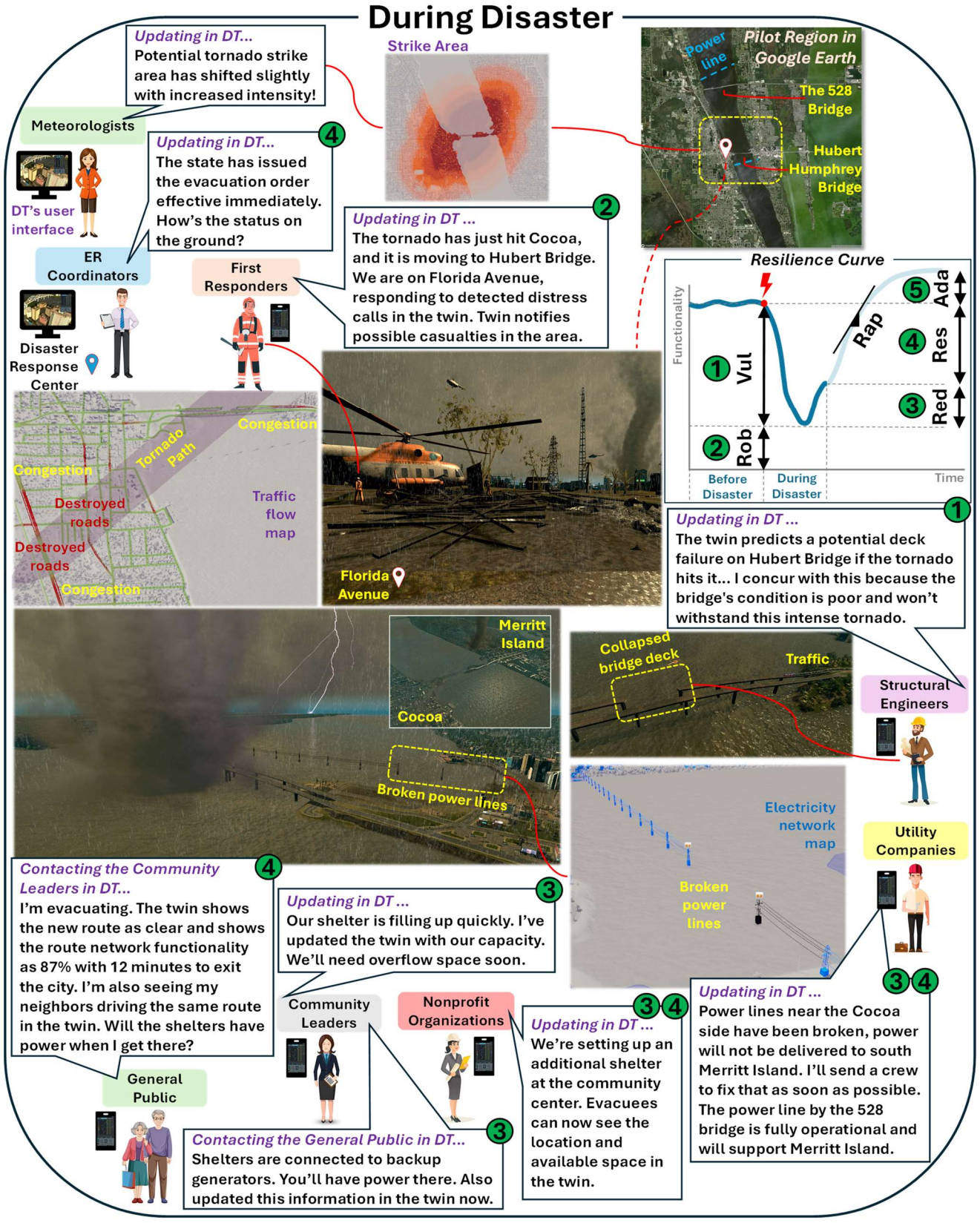


Figure 12. Collaborative use of DT during the tornado.

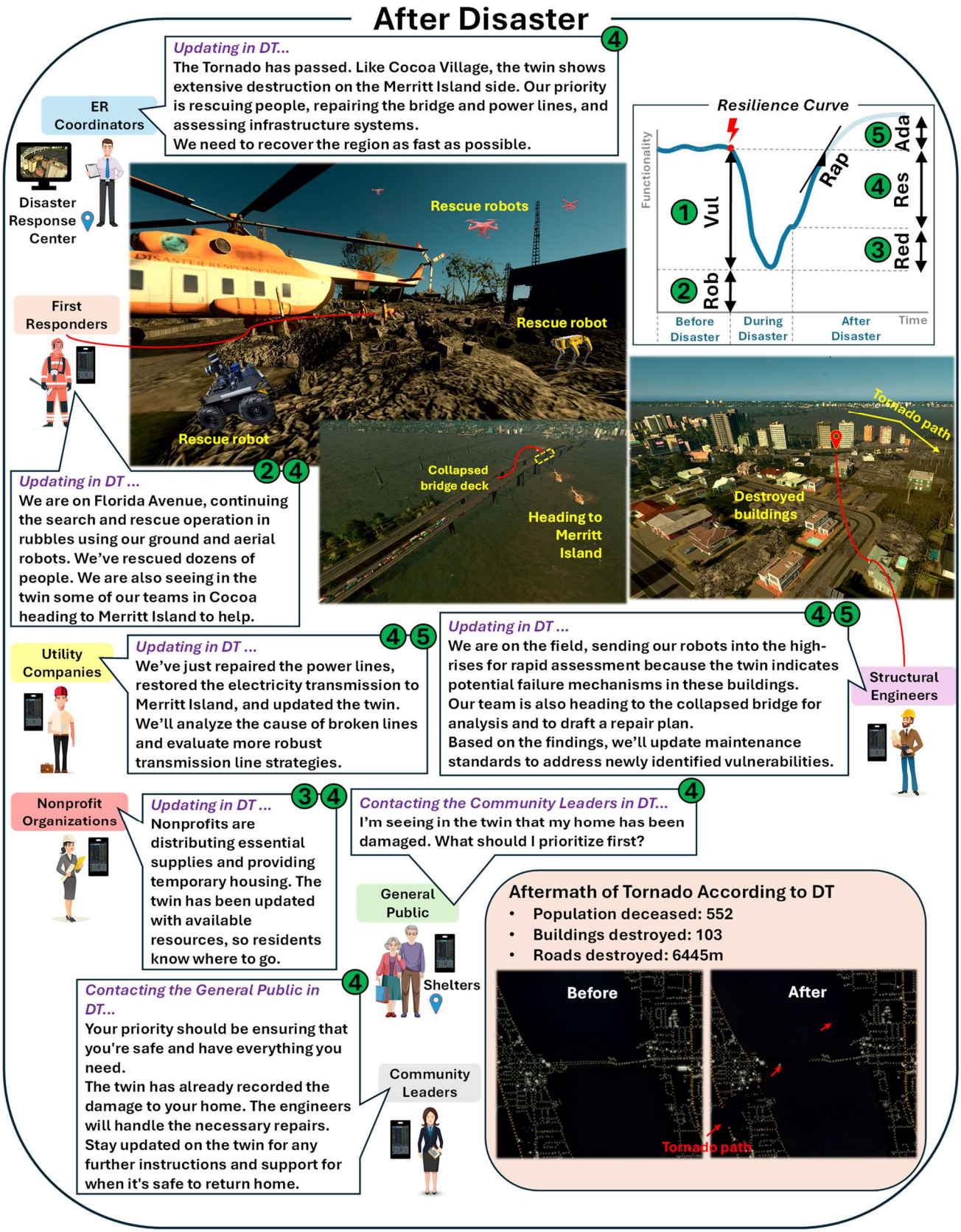


Figure 13. Collaborative use of DT after the tornado.

6.4. Additional Disaster Remarks

It is common for intense lightning to occur during a tornado, adding to the already severe and hazardous weather conditions, just as in real-world tornado disasters. When the tornado occurred in the DT, lightning strikes caused several locations in the pilot region to catch fire, though these were not considered in this paper for simplicity. Accounting for multiple or cascading disaster events alters the resilience curves in more intricate ways. Figure 14 presents some visuals of the wildfires that occurred during the tornado.

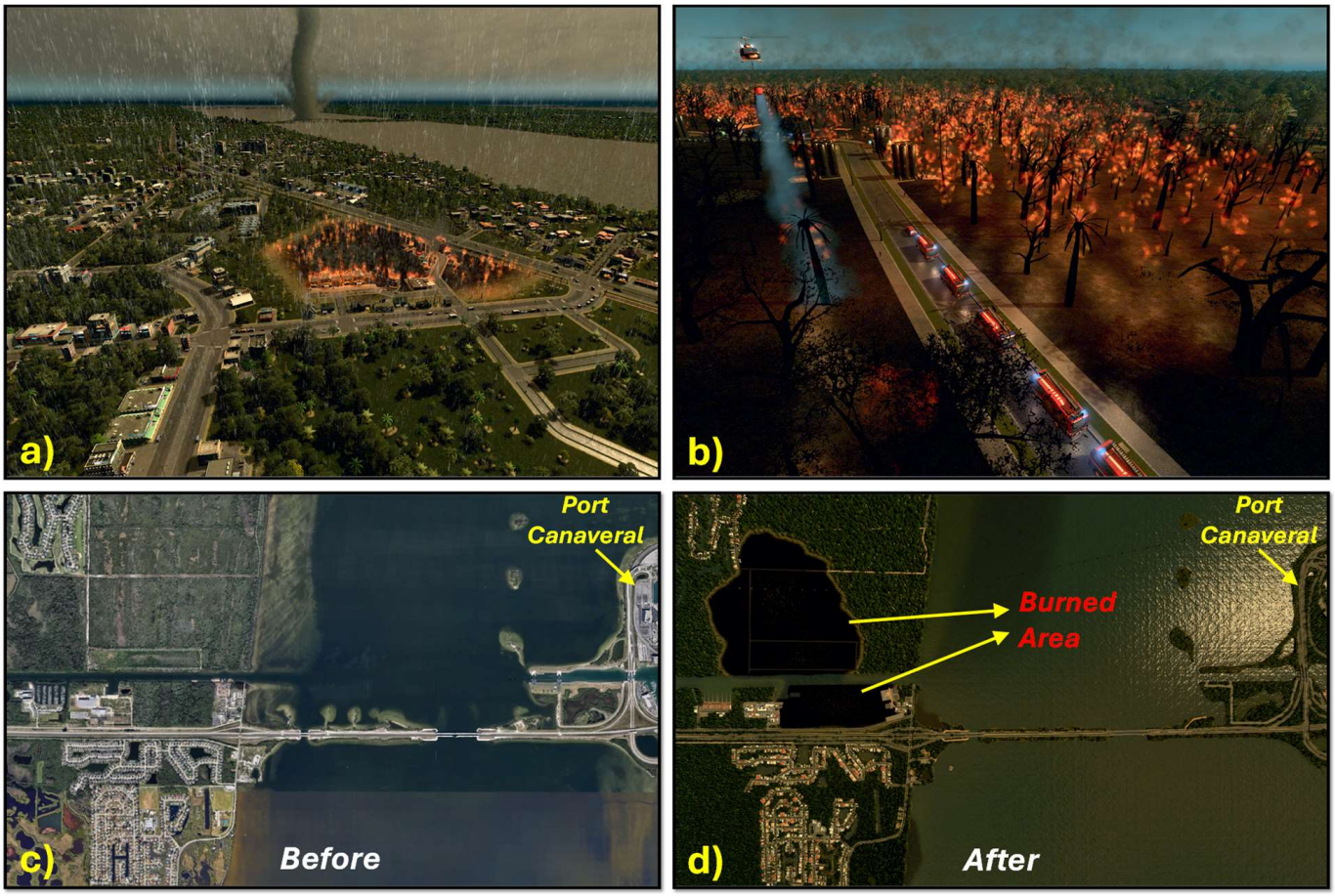


Figure 14. Wildfires caused by lightning strikes during the tornado in the pilot region: (**a**) Fire at the intersection of Eyster and Rockledge Boulevard during the strike of the tornado at the bridge; (**b**) Fire near SpaceX Rocket Assembly Site; (**c**) Before the fire map view in Google Earth; (**d**) After the fire map view in COWIN^{E}'s DT.

7. Research Directions with COWIN^{E}

The research directions using COWIN^{E} in the context of community disaster resilience are categorized under four general community dimensions, as illustrated in Figure 15: physical, social, environmental, and economic. Each dimension is further explored through specific research topics related to its respective challenges and opportunities. To visually align the research subjects with their respective categories, representative icons (most two related ones) for each community dimension are attached to the research topics. The following paragraphs provide a brief overview of several key research subjects associated with these dimensions.

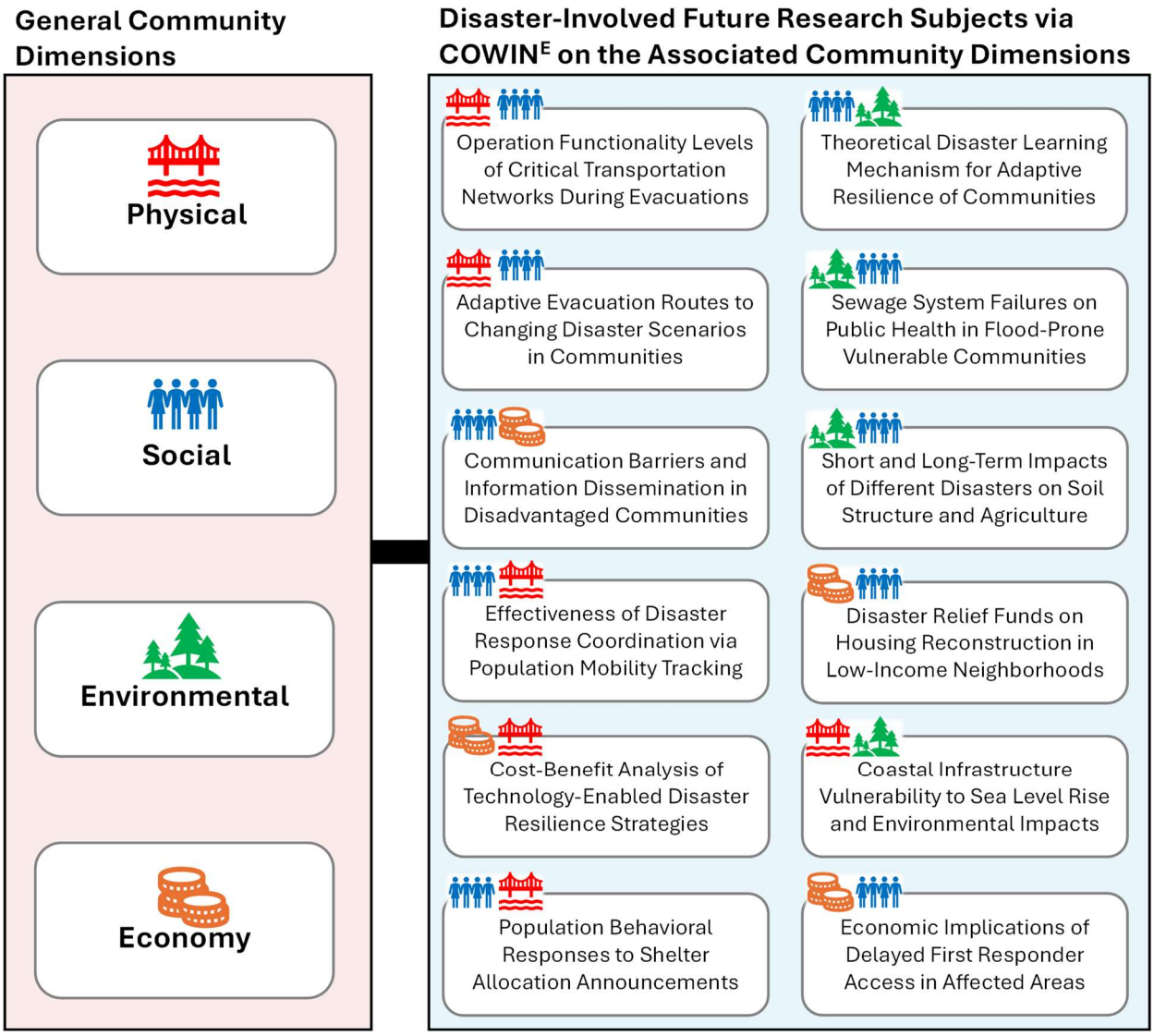


Figure 15. Future research subjects on community dimensions via COWIN^{E} (only two related community dimensions are shown in the research subjects).

The physical dimension focuses on the resilience and adaptability of infrastructure systems in the event of disasters. Research topics such as the operation functionality levels of critical transportation networks during evacuations highlight the importance of ensuring that key infrastructure, like roads and bridges, remains functional during emergencies. The study of adaptive evacuation routes to changing disaster scenarios in communities investigates how dynamic adjustments to routes can enhance safety and reduce congestion. These subjects explore the challenges posed by infrastructure vulnerabilities and the need for rapid adjustments in response to evolving disasters.

The social dimension investigates how communities interact with other community dimensions. The emphasis is placed on communication, information dissemination, and human behavior during disaster events. Research on communication barriers and the spread of information within disadvantaged and underserved communities addresses how gaps in communication can increase vulnerability before, during, and in the aftermath of disasters. The effectiveness of disaster response coordination via population mobility tracking analyzes how the real-time tracking of people's movements during emergencies can aid in resource allocation and improve response times. The study of population behavioral

responses to shelter allocation announcements provides insight into how communities react to critical information and how these behaviors can influence disaster outcomes.

The environmental dimension examines the complex interactions between natural ecosystems and disaster events. For instance, research into sewage system failures on public health in flood-prone vulnerable communities investigates how failing infrastructure during floods can exacerbate public health risks. Another subject addresses the short- and long-term impacts of different disasters on soil structure and agriculture, emphasizing the importance of understanding how disasters can disrupt food systems and long-term agricultural productivity. The vulnerability of coastal infrastructure to sea level rise and environmental impacts is a pressing issue for communities facing the dual threats of climate change and extreme weather events.

The economic dimension explores the financial implications of disaster resilience strategies. Topics such as the cost-benefit analysis of technology-enabled disaster resilience strategies examine how investing in advanced technologies can mitigate disaster impacts while considering the financial trade-offs. Disaster relief funds for housing reconstruction in low-income neighborhoods focuses on the economic disparities that arise after disasters, emphasizing the importance of equitable resource allocation. The economic implications of delayed first responder access in affected areas analyze how time-sensitive emergency responses can affect recovery costs, particularly in regions where infrastructure damage hinders rapid access.

Limitations

The following are some of COWIN^{E}'s known limitations. Subsequent studies will address these through the development and integration of modifications (mods) as part of C:S open accessibility.

- *Real-time data.* As mentioned earlier in the text, although the data used in COWIN^{E} are real-world data, the data flow mechanism is not real-time. COWIN^{E} relies on a manual data-integration process into its DT. Real-time data flow is an ideal mechanism in DTs, and it is a trending but challenging research subject.
- *Asset uniformity*. The elements in the real world are unique to each province in various aspects. For instance, supermarkets or hospitals in one location may appear similar but not identical to those in another location. However, in COWIN^{E}, these elements are generally uniform. This is due to COWIN^{E}'s use of a limited variety of standard physical assets to represent these real-world elements, though this can be addressed by developing and integrating additional mods.
- *Structural deterioration.* While the real-world elements in COWIN^{E} meet the basic functional requirements for their roles in cities (e.g., hospitals treat sick populations, contribute to population well-being, and operate within parameters like bed capacity), they lack comprehensive deterioration models to represent their structural conditions against external stressors over time. These elements are depicted in COWIN^{E} as either fully resistant or completely collapsing in the face of disasters (e.g., tornadoes or fires). With the deterioration models integrated, their remaining useful life could be tracked. This is the scope of the next study.
- *Traffic crashes.* Traffic crashes are an integral part of urban dynamics. Currently, COWIN^{E} does not feature a traffic crash mechanism, but the development and integration of such a mechanism are planned for future studies.
- *Fuel stations.* COWIN^{E} does not simulate fuel consumption, as the vehicles do not require fuel stations to operate. Fuel station mods are available on Steam, but they are only for aesthetic purposes. Addressing this is important as it would increase urban realism.
- *Data inconsistency.* During the integration of real-world data into COWIN^{E}, inconsistencies were observed between data sources related to different urban aspects, which was mentioned in the text earlier. This challenge may become more complex in the

later stages of DT development, especially when considering the implementation of a real-time data flow mechanism.

- *AV and EV Traffic Composition.* COWIN^{E} does not account for the dynamic composition of traffic of automated vehicles (AVs) and electric vehicles (EVs). It lacks mechanisms for assessing their impact on infrastructure and energy consumption. Further research and mod development are needed to include AV and EV interactions, along with their specific requirements, e.g., charging stations for EVs and the impact of AVs on traffic flow and patterns.

8. Conclusions and Final Remarks

This article introduces COWIN^{E} (Community Twin Ecosystem), which leverages Digital Twin (DT) technology to advance community disaster resilience. COWIN^{E} utilizes Cities:Skylines (C:S) as its base simulation engine and integrates real-world data to form its DT. COWIN^{E} is capable of capturing intricate community dynamics and interconnected structures, which aids in providing actionable insights for disaster planning. This ecosystem promotes collaboration in its utilization among decision-makers, stakeholders, and the general public for resilience planning before, during, and after disasters.

This paper identifies several limitations in the current landscape of community resilience research, including the utilization of static models that inadequately capture the complex, nonlinear responses of communities to disturbances rather than dynamic models. Additionally, many existing DT studies remain conceptual, focusing on high-level frameworks without demonstrating practical applications, while others concentrate on isolated dimensions, failing to account for the interconnected and multifaceted characteristics of communities. The visualization components in current DT applications are often static and lack interactivity, reducing their effectiveness for stakeholder engagement. By addressing these limitations, COWIN^{E} represents a considerable advancement in applying DT technology for community resilience. A case study conducted in Brevard County, Florida, demonstrates the application of COWIN^{E} in managing community resilience, particularly in the context of tornado scenarios. This collaborative engagement with COWIN^{E} highlights community vulnerabilities, adaptive strategies, and resource allocation essential for improving community resilience before, during, and after disasters. Moreover, this paper outlines future research directions across four key dimensions—physical, social, environmental, and economic—each with associated research topics that aim to explore further and validate the effectiveness of COWIN^{E}.

Importantly, the implications of COWIN^{E} extend beyond the United States and offer valuable insights for communities worldwide. As the frequency and intensity of disaster events increase due to climate change and urbanization, along with the growing complexities of urban environments, such as increased population density, infrastructure demands, and socioeconomic disparities, the challenges related to disaster management are becoming more pressing. Regions facing similar challenges can adopt COWIN^{E} to tailor their strategies according to local contexts. The flexibility of COWIN^{E} allows for adaptation to diverse geographical, social, and economic conditions, making it relevant for both developed and developing nations. For example, in earthquake-prone areas such as Turkey or Japan, COWIN^{E} can help local governments simulate potential earthquake impacts on urban infrastructure and develop tailored response strategies to minimize damage and enhance community safety. Similarly, a coastal city in Southeast Asia prone to typhoons could leverage COWIN^{E} to visualize flooding patterns and implement proactive evacuation strategies. As more people reside in high-risk urban areas, the need for effective disaster resilience strategies becomes imperative.

Overall, COWIN^{E}'s interactive and user-friendly interface enhances stakeholder engagement and decision-making capabilities, making it a valuable tool for real-time community resilience planning and response. This paper underscores the potential of COWIN^{E} to effectively address the challenges of implementing resilience practices in dis-

aster scenarios, paving the way for future empirical research to validate and refine this innovative framework.

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